COLLEGE ACCREDITATION EVIDENCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING
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BONAFIDE CERTIFICATE

Certified that this project report titled “College Accreditation Evidence Management System” is the bonafide work of **Nitin Aakash RK (240701370)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, **Nitin Aakash RK (240701370)**, hereby declare that the project report titled “College Accreditation Evidence Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report – including its data model, forms, reports, dashboard, Deluge scripts and workflows – was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
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NITIN AAKASH RK

ABSTRACT

Colleges preparing for accreditation by bodies such as NAAC and NBA must collect, organise and verify a large volume of documentary evidence – curriculum records, faculty development records, research outputs, student achievement and scholarship records and many more – across departments and academic years. In most institutions this evidence is gathered through e-mail, shared folders, pen drives, paper files and spreadsheets. As a result, documents are scattered, inconsistently classified and difficult to verify, and the accreditation cell has no live view of how much evidence is pending, approved or rejected.

The College Accreditation Evidence Management System addresses this problem with a centralised application built on Zoho Creator, a low-code platform. The application contains six forms – Department, Faculty, Accreditation Criteria, Academic Year, Evidence Submission and Verification – linked through six lookup relationships, six list reports, and an Accreditation_Dashboard with four count cards and three charts computed with aggregate count functions. Form workflows send e-mail notifications through the Deluge sendmail task, and Deluge logic on the Academic_Year_Form closes a year whose end date has passed and sends a deadline reminder when the record is saved.

The application was examined using sixteen screenshots of the working system. Forms, lookups, reports and dashboard values are evidenced by these screenshots, and e-mail delivery from the notification workflows was confirmed by the developer. Tests whose results were not yet recorded are clearly marked. The study also identified limitations of the current build – on-save rather than scheduled automation, a fixed notification recipient, an unrenamed reviewer field, selectable inactive departments and incomplete sample data – which form the basis of the future enhancements proposed. The project shows that a low-code platform can deliver a practical, maintainable evidence management system for accreditation with very little code.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Accreditation; NAAC; Evidence Management

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Accreditation Evidence Management System
Application Domain	Education – accreditation evidence management (NAAC-style)
Student Name & Register No.	Nitin Aakash RK – 240701370
Zoho Creator Edition	Trial Edition (as shown in the application header)
Application Name	College Accreditation Evidence Management System
Live URL	creatorapp.zoho.in/nitinaakash.rk.2024.cse_r965/college-accreditation-evidence-management-system
Application Account	nitinaakash.rk.2024.cse@rajalakshmi.edu.in
Number of Forms / Subforms	6 / 0 (Department_Form, Faculty_Form, Accreditation_Criteria_Form, Academic_Year_Form, Evidence_Submission_Form, Verification_Form)
Lookup Relationships	6
Number of Reports	6 list reports (one per form); filtered reports not built
Pages / Dashboards	1 – Accreditation_Dashboard (4 COUNT cards, 1 donut chart, 2 bar charts)
Workflows	Form workflows on 4 forms: WF5, WF6 evidenced by screenshots; WF1–WF3 e-mail delivery developer-confirmed; WF4 awaiting evidence. Schedules: none
Deluge Scripts Captured	2 (WF5 complete; WF6 lines 1–10)
Integrations	E-mail through Deluge sendmail
Roles	Recorded as Role values in Faculty_Form; permission profiles not confirmed
Environments	Single live application; separate environments not evidenced

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CSE	Computer Science and Engineering
DS	Deluge Script
ECE	Electronics and Communication Engineering
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
HTTPS	HyperText Transfer Protocol Secure
ID	Identifier
IQAC	Internal Quality Assurance Cell
KPI	Key Performance Indicator
NAAC	National Assessment and Accreditation Council
NBA	National Board of Accreditation
NFR	Non-Functional Requirement
PK	Primary Key
REST	Representational State Transfer
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
WF	Workflow
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Accreditation has become one of the most important quality-assurance activities in Indian higher education. Bodies such as the National Assessment and Accreditation Council (NAAC) [1] and the National Board of Accreditation (NBA) [2] assess an institution or a programme against a published framework of criteria, key indicators and metrics. For every metric the institution is expected to produce documentary evidence – circulars, minutes of meetings, faculty development certificates, research publications, student achievement records, scholarship lists, feedback analyses and many other records – and to present them in an organised form during the self-study and peer-team visit.

In a typical engineering college, this evidence is generated across many departments and over several academic years. Individual faculty members create most of the documents, department coordinators collect them, and an institution-level accreditation cell (often the Internal Quality Assurance Cell) consolidates them for the final submission. Because the documents originate in different places and at different times, the collection process usually depends on e-mail threads, shared drives, pen drives, printed files and spreadsheets maintained by each coordinator.

This project, the **College Accreditation Evidence Management System**, is a low-code application built on Zoho Creator [4] to centralise this activity. It provides forms to maintain departments, faculty, accreditation criteria and academic years; a form through which evidence is submitted with an attached document; a form through which a reviewer records a verification decision; list reports for every form; e-mail notifications driven by form workflows; and an Accreditation_Dashboard that summarises evidence and verification counts. The department names, faculty names, criteria and records used in this report are sample data entered for development and demonstration.

1.2 Problem Statement

Colleges preparing for accreditation do not have a single, structured record of the evidence that has been submitted for each criterion, by which department, for which academic year, and whether it has been checked. Evidence is scattered across personal mailboxes and folders, its verification status is tracked informally, and the accreditation cell cannot see progress without repeatedly asking coordinators. The specific problems are:

- **Scattered storage:** documents are kept in e-mail attachments, local folders and paper files, so the same document may exist in several versions or may be lost.
- **No consistent classification:** evidence is not uniformly tagged with its department, accreditation criterion and academic year, which makes criterion-wise compilation slow.
- **Informal verification:** whether a document has been reviewed, accepted or rejected is recorded, if at all, in personal notes or spreadsheets, with no reviewer comments or review date.
- **Poor visibility of progress:** the number of pending, approved and rejected items is not available without manual counting.
- **Missed deadlines:** submission deadlines for each academic year are communicated manually, and reminders depend on individual effort.

1.3 Objectives

The objectives of the project are:

- To design a data model and six Zoho Creator forms for departments, faculty, accreditation criteria, academic years, evidence submissions and evidence verification.
- To link evidence to its department, criterion and academic year, and verification to its evidence and reviewer, using lookup fields.
- To provide list reports through which every category of record can be viewed, searched and maintained.
- To automate selected actions – e-mail notifications, academic-year status update and a deadline reminder – using form workflows and Deluge scripts.
- To provide a dashboard that shows total, pending, approved and rejected evidence counts and charts of evidence by status, department and academic year.
- To reduce paperwork and improve the accessibility and organisation of accreditation records.

1.4 Scope of the Project

In scope. The application covers the maintenance of master data (departments, faculty, accreditation criteria and academic years), the submission of evidence records with an attached document, the recording of a verification decision (Pending, Approved or Rejected) with comments and a review date, e-mail notifications through form workflows, on-save Deluge logic on the Academic_Year_Form, six list reports and one dashboard page. The criteria data entered in the application follows the NAAC

framework; the Framework field also allows NBA and Other, so the same structure can be adapted to other frameworks.

Out of scope in the present version. The application does not include scheduled (time-triggered) workflows, filtered reports such as Pending_Verification_Report, role-based permission profiles enforced by Zoho Creator, a customer portal, approval processes or Blueprints, multiple deployment environments, external API integrations, AI features or a tested mobile deployment. These items are discussed as future enhancements in Chapter 9.

1.5 Justification for a Low-Code Approach

Low-code application platforms allow applications to be built through visual modelling and a small amount of scripting rather than hand-written code for every layer [3]. Zoho Creator provides forms, reports, pages, workflows, hosting and security as platform services, and custom logic is written in the Deluge scripting language [10]. Industry analysts have reported rapid adoption of such platforms for departmental and process applications [5]. For an accreditation evidence system, a low-code approach is suitable for the following reasons:

- **Speed of development:** forms, lookups, reports and a dashboard can be assembled visually, leaving time to concentrate on the data model and workflows.
- **Maintainability by college staff:** a new department, criterion or academic year is added as a record, not as a code change.
- **Built-in hosting and storage:** the application and its uploaded files are hosted on Zoho's cloud without a separate server or database.
- **Native automation:** form workflows and the Deluge sendmail task provide notifications without an external mail server.
- **Low cost for a pilot:** the application was built on a Zoho Creator trial edition.

1.6 Organisation of the Report

Chapter 2 studies the existing system and reviews related literature on low-code development and accreditation frameworks. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design, including the architecture, data model, form design and process flow. Chapter 5 explains the implementation on Zoho Creator with screenshots of the actual application. Chapter 6 covers testing, clearly separating evidenced results from results still to be confirmed. Chapter 7 discusses hosting, security and governance. Chapter 8 presents the results, the achievement of objectives and the observed limitations, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system, evidence for accreditation is collected through a largely manual, coordinator-driven process. When the accreditation cell prepares for a self-study report, it circulates a list of criteria and the evidence required for each. Faculty members locate or prepare the relevant documents and send them to their department coordinator by e-mail, on a pen drive or as printed copies. The coordinator saves the files in local folders or records them in a spreadsheet, and forwards the department's bundle to the accreditation cell. The cell checks each document; if anything is missing, incorrect or in the wrong format, it follows up by telephone or e-mail and the faculty member resubmits. The cycle repeats until the evidence for every criterion is complete. Figure 2.1 shows this as-is process.

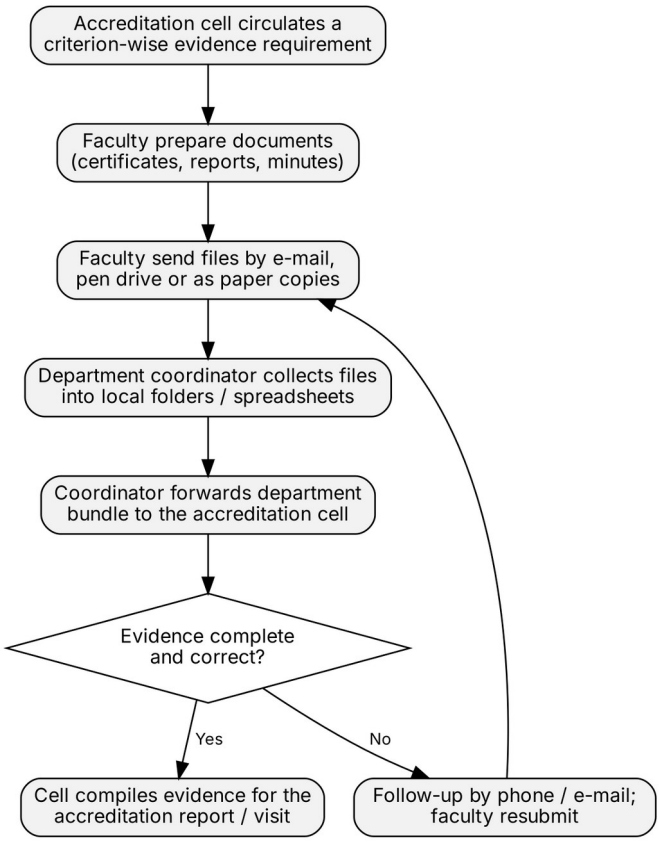


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Documents are distributed across mailboxes, local drives and paper files, so there is no single authoritative copy.
- Evidence is not consistently classified by department, criterion and academic year, so criterion-wise compilation needs manual sorting.

- Verification decisions, reviewer comments and review dates are not recorded in a structured way.
- The accreditation cell has no live view of how much evidence is pending, approved or rejected.
- Deadline reminders and status updates depend on individual follow-up rather than on the system.
- When coordinators change, the knowledge of where evidence is stored is easily lost.

2.3 Low-Code versus Traditional Development

Table 2.1 compares building the proposed application with a traditional pro-code stack (for example, a Java or Python web framework with a relational database) and with Zoho Creator. The time figures are planning estimates for an application of this size, not measured values.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time (estimate)	Several weeks for back end, front end, database and testing	Days to a few weeks for forms, reports, workflows and dashboard
Skills required	Full-stack programming, database design, deployment	Data modelling, form and report design, basic Deluge scripting
Hosting and storage	Server or cloud instance, database and file storage to be arranged	Hosted on Zoho cloud; uploaded files stored by the platform
Relationships between data	Foreign keys and joins coded manually	Lookup fields configured in the form builder
Notifications	Mail server or e-mail API integration to be coded	Deluge sendmail task inside a form workflow
Dashboards	Charting library and queries to be coded	Page with aggregate (count) widgets and charts
Customisation limits	Practically unlimited	Bounded by platform features and plan limits
Cost	Developer effort plus hosting cost	Subscription (trial edition used for this project)

2.4 Literature Survey

The survey covered academic studies of low-code platforms, the official accreditation manuals that define what evidence an institution must maintain, and the Zoho Creator and Deluge documentation used for implementation. Table 2.2 summarises the sources.

Table 2.2: Summary of Literature Survey

No.	Author(s) / Source	Title / Product	Key Findings	Relevance to this Project
1	R. Sanchis et al., 2020 [6]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Low-code allows domain users to build and adapt process applications quickly with less dependence on specialist developers.	Supports building the evidence system so that college staff can maintain it.
2	A. C. Bock and U. Frank, 2021 [7]	Low-Code Platform	Defines low-code platforms by visual modelling, generated runtime and managed deployment, and notes the risk of vendor lock-in.	Frames the platform choice and the limitations discussed in Section 8.5.
3	D. Di Ruscio et al., 2022 [8]	Low-Code Development and Model-Driven Engineering	Places low-code within model-driven engineering, where applications are derived from data and process models.	Justifies designing the ER model before the forms (Chapter 4).
4	Y. Luo et al., 2021 [9]	Characteristics and Challenges of Low-Code Development	Practitioners value speed but report limits in customisation, debugging and documentation.	Informs the challenges recorded in Section 6.4.
5	NAAC [1]	Institutional accreditation manuals	Defines criteria, key indicators and metrics for which institutions must keep documentary evidence.	Basis for the Accreditation_Criteria _Form and sample criteria.
6	NBA [2]	Programme accreditation manuals	Defines programme-level criteria with their own evidence requirements.	Reason for including NBA as a Framework option.
7	Zoho Corporation [4], [10]	Zoho Creator Help and Deluge Reference	Describes forms, lookups, reports, pages, workflows, schedules and the sendmail task.	Primary technical reference for Chapter 5.

2.5 Overview of Accreditation Frameworks

NAAC assesses institutions against seven criteria, each divided into key indicators and metrics [1]. NBA accredits individual programmes, mainly in technical education, against its own set of criteria [2]. Both frameworks require evidence to be organised criterion-wise and year-wise, which is why the proposed data model treats Framework,

Criterion_Number and Academic_Year as first-class records. Table 2.3 lists the seven NAAC criteria for reference.

Table 2.3: NAAC Criteria Used as the Reference Framework

Criterion	Title	Typical Evidence
1	Curricular Aspects	Curriculum documents, feedback records, value-added course details
2	Teaching-Learning and Evaluation	Teaching plans, faculty development records, result analyses
3	Research, Innovations and Extension	Publications, funded projects, extension activity reports
4	Infrastructure and Learning Resources	Facility records, library and IT resource details
5	Student Support and Progression	Scholarship records, placement and achievement records
6	Governance, Leadership and Management	Policies, meeting minutes, welfare measures
7	Institutional Values and Best Practices	Best-practice reports, environmental and gender initiatives

The application currently holds three NAAC sample criteria – 1.1 Curriculum Planning, 2.3 Teaching and Learning and 3.2 Research and Innovation (Figure 5.9). These are short working names entered for demonstration and are not reproduced from the official metric titles.

2.6 Emerging Trends in Low-Code Development

- **AI-assisted building:** platforms now include AI assistants; the Zoho Creator builder used in this project shows the Zia “Build Agent” panel beside the workflow editor (Figure 5.14), although it was not relied on for the features described in this report.
- **Citizen development:** process owners, such as accreditation coordinators, increasingly build or adapt their own applications.
- **Automation-first design:** notifications, reminders and status changes are expected to be handled by workflows instead of manual follow-up.
- **Multi-device access:** low-code applications are expected to work on web, tablet and mobile from one definition.

2.7 Proposed System

The proposed system replaces scattered files and spreadsheets with one Zoho Creator application. Master data for departments, faculty, criteria and academic years is maintained in dedicated forms. Faculty submit evidence through the

Evidence_Submission_Form, choosing the department, criterion and academic year from lookups and attaching the document. A reviewer records the decision in the Verification_Form. Form workflows send e-mail notifications and apply simple rules to academic-year records, and the Accreditation_Dashboard gives a summary of evidence and verification status. Because every record carries its department, criterion and academic year, evidence can be compiled criterion-wise and year-wise from the reports.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and Role Values

Four kinds of users are involved. In the present version these are recorded as values of the Role dropdown in the Faculty_Form (Faculty, Department Coordinator, Accreditation Admin) together with the person who performs verification (Reviewer). They describe each person's responsibility; separate Zoho Creator permission profiles for these roles are not confirmed, so all screens described in this report were used through the application owner's account.

Table 3.1: Stakeholders and Role Values

Stakeholder	Recorded As	Responsibilities
Faculty	Role = Faculty	Submit evidence records with the supporting document
Department Coordinator	Role = Department Coordinator; Coordinator_Email in Department_Form	Monitor the department's submissions; receive notifications
Reviewer	Faculty record selected in the reviewer lookup of Verification_Form	Record the decision, comments and review date for evidence
Accreditation Admin	Role = Accreditation Admin	Maintain departments, faculty, criteria and academic years; monitor the dashboard

3.2 Functional Requirements

Table 3.2 lists the functional requirements and the status of each in the current application. "Met" means the requirement is visible in the screenshots or confirmed by the developer; "Partially met" means it works with a limitation noted; "Future" means it is planned but not implemented.

Table 3.2: Functional Requirements

FR ID	Requirement	Status
FR-01	Maintain departments with name, code, coordinator e-mail and Active/Inactive status	Met
FR-02	Maintain faculty with name, e-mail, department (lookup), role and status	Met
FR-03	Maintain accreditation criteria with framework, number, name, description and priority	Met (descriptions not yet entered)
FR-04	Maintain academic years with start, end and submission-deadline dates and status	Met

FR ID	Requirement	Status
FR-05	Submit evidence linked to department, criterion and academic year, with a document upload	Met (no files attached in sample records)
FR-06	Record a verification decision (Pending/Approved/Rejected) with reviewer, comments and date	Met (reviewer field labelled “Faculty_Form”)
FR-07	View, search and maintain all records through list reports	Met (six default reports)
FR-08	Send e-mail notifications for new evidence, verification decisions and new faculty	Met – developer-confirmed delivery
FR-09	Mark an academic year Closed when its end date has passed	Partially met (on save only)
FR-10	Remind about an approaching submission deadline	Partially met (on save only)
FR-11	Show total, pending, approved and rejected counts and charts on a dashboard	Met
FR-12	Provide filtered reports for active records, pending verification and rejected evidence	Future
FR-13	Allow only active departments to be selected for new evidence	Future
FR-14	Restrict each role to its own data through permission profiles	Future

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms use dropdowns, radio buttons, lookups and date pickers so that users choose values instead of typing them.
NFR-02	Availability	The application is hosted on Zoho’s cloud and accessed through a web browser without local installation.
NFR-03	Data integrity	Lookup fields keep evidence and verification records linked to valid master records.
NFR-04	Maintainability	New departments, criteria and academic years are added as data, without changing the application.
NFR-05	Scalability	The same structure supports more departments, criteria and years and other frameworks (NBA, Other).
NFR-06	Security	Access requires a Zoho sign-in; data is held on the Zoho platform (Section 7.3).
NFR-07	Traceability	Every record has an auto-generated ID, and verification records carry the reviewer and review date.

3.4 Hardware and Software Requirements

Table 3.4 lists the environment used for development. Hardware details were not available at the time of writing and are left as placeholders to be confirmed.

Table 3.4: Hardware and Software Requirements

Processor	[CONFIRM PROCESSOR]
Memory (RAM)	[CONFIRM RAM]
System type	[CONFIRM SYSTEM TYPE]
Operating system	Windows 11 (as seen in the screenshots; subject to confirmation)
Web browser	Google Chrome (as seen in the screenshots; subject to confirmation)
Development platform	Zoho Creator (Trial Edition)
Scripting language	Deluge
E-mail	Deluge sendmail from the Zoho admin user address
Network	Internet connection to creator.zoho.in and creatorapp.zoho.in

3.5 Use Case Model

Figure 3.1 shows the use cases of the four actors. Faculty submit and view evidence; the Department Coordinator monitors submissions and the dashboard; the Reviewer records verification decisions; and the Accreditation Admin maintains the master data and monitors overall progress. All actors may receive e-mail notifications. Tables 3.5 to 3.7 describe the three main use cases.

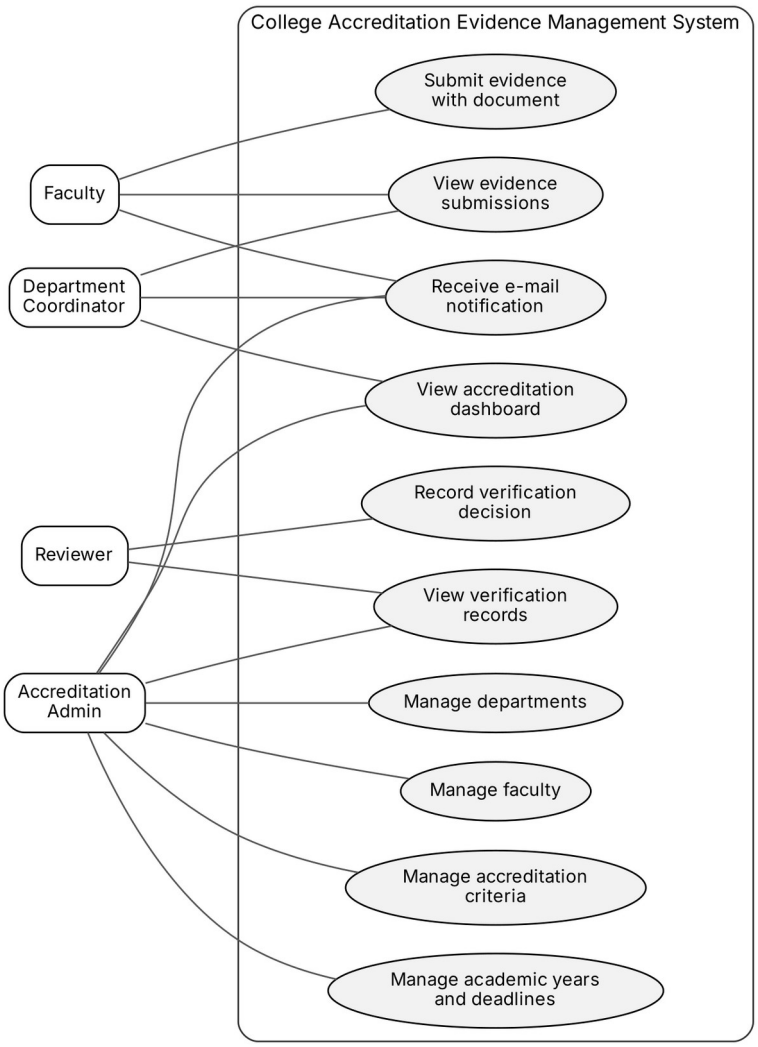


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Submit Evidence

Use Case ID	UC-01
Use Case Name	Submit Evidence
Actor(s)	Faculty
Pre-condition	The department, criterion and academic year exist as records.
Main Flow	1. The user opens Evidence_Submission_Form. 2. Enters the Evidence_Title. 3. Selects the Department, Criterion and Academic_Year from lookups. 4. Attaches the Evidence_Document. 5. Clicks Submit; the record is saved with an auto-generated Evidence_ID. 6. The form workflow sends an e-mail notification.
Alternate Flow	The user submits without attaching a file; the record is still saved because the upload field is not mandatory (observed in the sample data).
Post-condition	The evidence appears in Evidence_Submission_Form Report and in the dashboard counts and charts.
Related FR	FR-05, FR-08, FR-11

Table 3.6: Use Case Description – Verify Evidence

Use Case ID	UC-02
Use Case Name	Verify Evidence
Actor(s)	Reviewer
Pre-condition	The evidence has been submitted and the reviewer exists in Faculty_Form.
Main Flow	1. The reviewer opens Verification_Form. 2. Selects the Evidence (shown by Evidence_Title). 3. Selects the reviewer's own faculty record. 4. Chooses Pending, Approved or Rejected. 5. Enters Review_Comments and Review_Date. 6. Clicks Submit; the decision e-mail workflow runs.
Alternate Flow	If further documents are needed, the reviewer keeps the decision Pending and records the reason in the comments.
Post-condition	The decision appears in Verification_Form Report and the dashboard counts change accordingly.
Related FR	FR-06, FR-08, FR-11

Table 3.7: Use Case Description – Monitor Accreditation Progress

Use Case ID	UC-03
Use Case Name	Monitor Accreditation Progress
Actor(s)	Accreditation Admin, Department Coordinator
Pre-condition	Evidence and verification records exist.
Main Flow	1. The user opens Accreditation_Dashboard from the navigation menu. 2. Reads the four count cards. 3. Reads the status, department-wise and academic-year-wise charts. 4. Opens the relevant report for record-level detail.
Alternate Flow	If no records exist, the counts show zero and the charts are empty.
Post-condition	The user knows the current position of evidence and verification.
Related FR	FR-07, FR-11

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application follows the three-layer structure that Zoho Creator provides (Figure 4.1). The **presentation layer** consists of the six forms used for data entry, the six list reports used to view and maintain records, and the Accreditation_Dashboard page. The **logic layer** consists of form workflows that run when a record is created or edited, the Deluge scripts inside those workflows, and the aggregate (count) functions that compute the dashboard figures. The **data layer** holds the form records, the lookup relationships between them and the files uploaded in the Evidence_Document field. E-mail notifications leave the application through the Deluge sendmail task, using the Zoho admin user address as the sender. Users reach the application through a web browser.

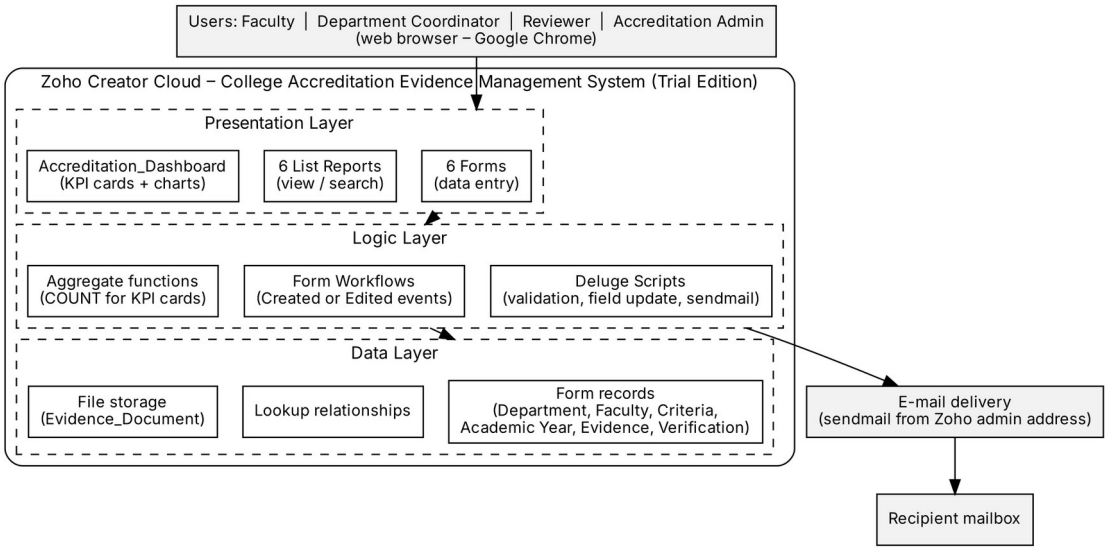


Figure 4.1: System Architecture of the Application

4.2 Data Model

The data model was designed before the forms were built. It contains four master entities – Department, Faculty, Accreditation Criteria and Academic Year – and two transaction entities – Evidence Submission and Verification. Each entity is a Zoho Creator form; each relationship is a lookup field in the child form that points to the parent form and displays a readable field of the parent. Figure 4.2 shows the entity-relationship diagram and Table 4.1 summarises the relationships.

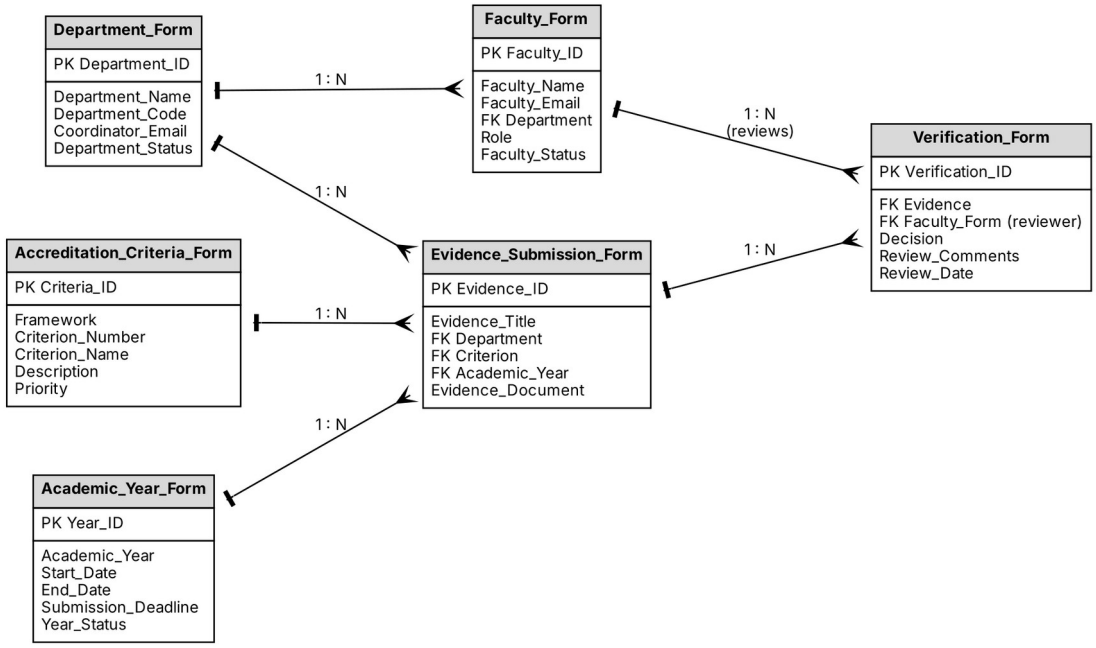


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Parent Entity	Child Entity	Lookup Field (in Child)	Displayed Value	Cardinality
Department_Form	Faculty_Form	Department	Department_Name	1 : N
Department_Form	Evidence_Submission_Form	Department	Department_Name	1 : N
Accreditation_Criteria_Form	Evidence_Submission_Form	Criterion	Criterion_Name	1 : N
Academic_Year_Form	Evidence_Submission_Form	Academic_Year	Academic_Year	1 : N
Evidence_Submission_Form	Verification_Form	Evidence	Evidence_Title	1 : N
Faculty_Form	Verification_Form	Faculty_Form (reviewer)	Faculty_Name	1 : N

A one-to-many relationship between evidence and verification allows the same evidence to be reviewed more than once, for example a Pending review followed later by an Approved review. The status of evidence is therefore derived from its verification records rather than stored on the evidence itself.

4.3 Form Design

Each form was limited to at most six fields so that data entry stays quick and focused. Tables 4.2 to 4.7 give the field specification of each form. The “Required” column reflects what is visible in the live forms: a red asterisk marks a mandatory field. Auto-

number ID fields do not appear on the data-entry screen; their values are visible in the reports.

Table 4.2: Field Specification – Department_Form

Field Label	Link Name	Field Type	Required	Remarks
Department ID	Department_ID	Auto Number	—	System generated (1, 2, 3 …)
Department Name	Department_Name	Single Line	No	e.g., Computer Science and Engineering; shown in lookups
Department Code	Department_Code	Single Line	No	e.g., CSE, ECE, CIVIL
Coordinator Email	Coordinator_Email	Email	No	Displayed as a mail link in the report
Department Status	Department_Status	Radio	No	Active / Inactive

Table 4.3: Field Specification – Faculty_Form

Field Label	Link Name	Field Type	Required	Remarks
Faculty ID	Faculty_ID	Auto Number	—	System generated
Faculty Name	Faculty_Name	Single Line	Yes	Shown in the reviewer lookup
Faculty Email	Faculty_Email	Email	Yes	Displayed as a mail link
Department	Department	Lookup	No	Department_Form, shows Department_Name
Role	Role	Dropdown	No	Faculty / Department Coordinator / Accreditation Admin
Faculty Status	Faculty_Status	Radio	No	Active / Inactive

Table 4.4: Field Specification – Accreditation_Criteria_Form

Field Label	Link Name	Field Type	Required	Remarks
Criteria ID	Criteria_ID	Auto Number	—	System generated
Framework	Framework	Dropdown	No	NAAC / NBA / Other
Criterion Number	Criterion_Number	Single Line	No	e.g., 1.1, 2.3, 3.2
Criterion Name	Criterion_Name	Single Line	No	Shown in the evidence lookup
Description	Description	Multi Line	No	Empty in the current sample records
Priority	Priority	Radio	No	High / Medium / Low

Table 4.5: Field Specification – Academic_Year_Form

Field Label	Link Name	Field Type	Required	Remarks
Year ID	Year_ID	Auto Number	—	System generated
Academic Year	Academic_Year	Single Line	No	e.g., 2026–2027; shown in the evidence lookup
Start Date	Start_Date	Date	No	Format dd-MMM-yyyy
End Date	End_Date	Date	No	Used by WF5
Submission Deadline	Submission_Deadline	Date	No	Used by WF6
Year Status	Year_Status	Radio	No	Active / Closed

Table 4.6: Field Specification – Evidence_Submission_Form

Field Label	Link Name	Field Type	Required	Remarks
Evidence ID	Evidence_ID	Auto Number	—	System generated
Evidence Title	Evidence_Title	Single Line	No	Shown in the verification lookup
Department	Department	Lookup	No	Department_Form; inactive departments are also listed
Criterion	Criterion	Lookup	No	Accreditation_Criteria_Form, shows Criterion_Name
Academic Year	Academic_Year	Lookup	No	Academic_Year_Form, shows Academic_Year
Evidence Document	Evidence_Document	File Upload	No	No file attached in the sample records

Table 4.7: Field Specification – Verification_Form

Field Label	Link Name	Field Type	Required	Remarks
Verification ID	Verification_ID	Auto Number	—	System generated
Evidence	Evidence	Lookup	No	Evidence_Submission_Form, shows Evidence_Title
Faculty_Form (Reviewer)	Faculty_Form	Lookup	No	Faculty_Form, shows Faculty_Name; label not yet renamed to “Reviewer”
Decision	Decision	Radio	No	Pending / Approved / Rejected

Field Label	Link Name	Field Type	Required	Remarks
Review Comments	Review_Comments	Multi Line	No	Free-text remarks
Review Date	Review_Date	Date	No	Format dd-MMM-yyyy

4.4 Process and Workflow Design

Figure 4.3 shows the life cycle of an evidence item. The Accreditation Admin first maintains the master records. A faculty member then submits evidence, which triggers an e-mail notification. A reviewer records a decision in the Verification_Form; the decision triggers a second e-mail. A Rejected item is followed by a fresh submission, and a Pending item is reviewed again later. Every saved record is immediately reflected in the reports and the dashboard.

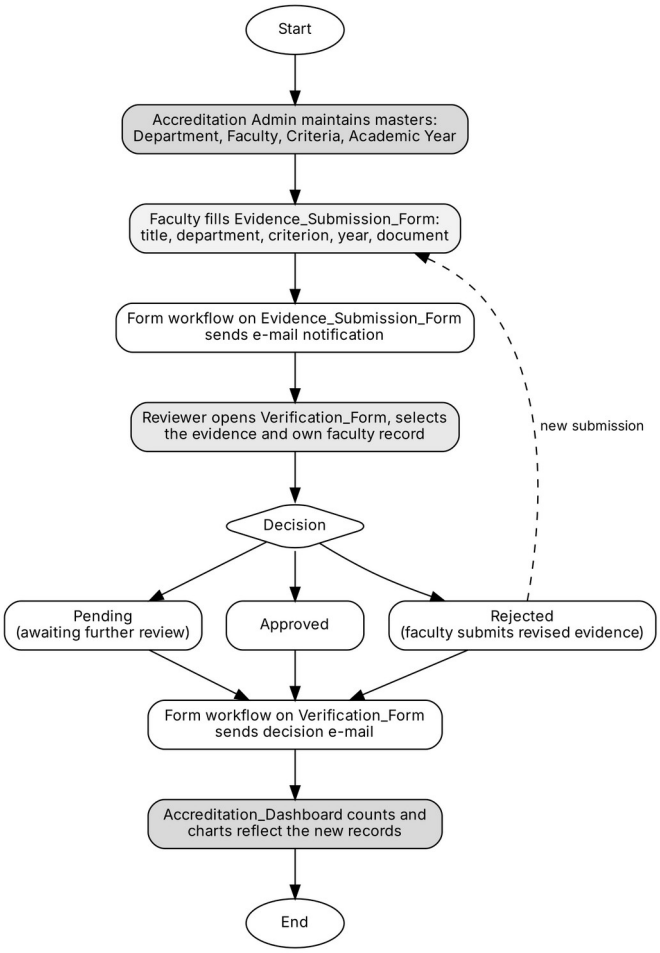


Figure 4.3: Process Flow of an Evidence Item

Six workflows were planned for the application. Table 4.8 lists the design of each. Their implementation status, which differs between workflows, is given in Section 5.6.

Table 4.8: Planned Workflow Design

WF	Name	Form	Planned Trigger	Planned Action
WF1	New Evidence Submission Notification	Evidence_Submission_Form	On submission	E-mail about the new evidence
WF2	Verification Decision Notification	Verification_Form	On submission	E-mail with the decision and comments
WF3	New Faculty Registration Notification	Faculty_Form	On submission	E-mail about the newly registered faculty
WF4	Default Verification Decision	Verification_Form	On load / submission	Set Decision to Pending when appropriate
WF5	Academic Year Closure	Academic_Year_Form	Record created or edited	Set Year_Status to Closed after End_Date
WF6	Upcoming Submission Deadline Reminder	Academic_Year_Form	Record created or edited	E-mail when the deadline is 0–7 days away

4.5 User Interface and Navigation Design

The application uses Zoho Creator's left-side navigation menu (Figure 4.4). Each of the six forms is a menu section that expands to two items: the form itself, for entering a new record, and its report, for viewing existing records. The seventh menu item opens the Accreditation_Dashboard page. Section names follow the form link names so that the menu, reports and scripts use one consistent vocabulary. The header bar provides the search, add-record and more-actions buttons on reports, and preview icons for web, tablet and mobile layouts.

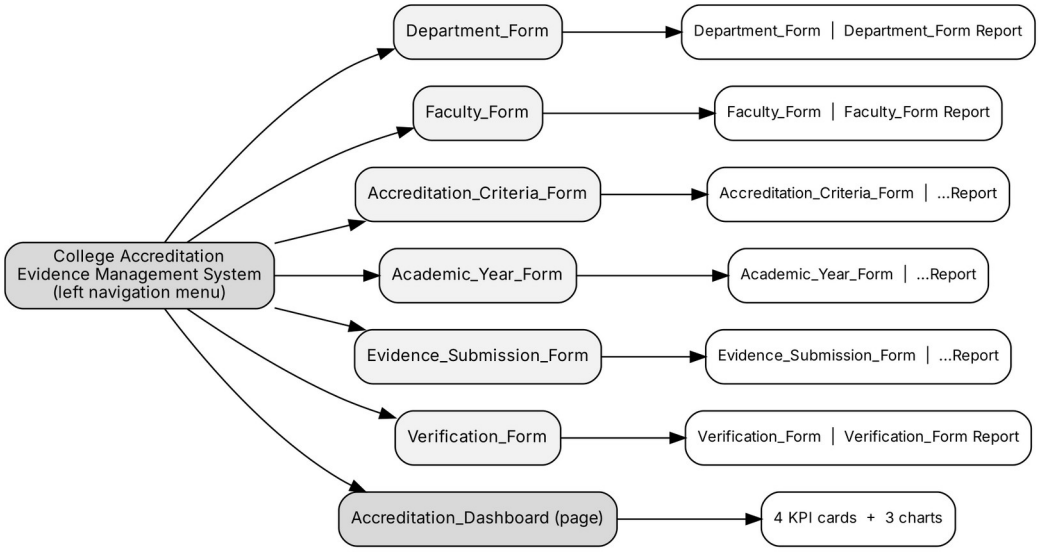


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Access in the current version is through the application owner's Zoho account; separate permission profiles for the four roles are not confirmed. Table 4.9 therefore presents a **proposed** role–permission matrix that can be configured in Zoho Creator's user and permission settings in a future version. It is a design recommendation, not a description of current behaviour.

Table 4.9: Proposed Role–Permission Matrix (Not Yet Configured)

Form / Page	Faculty	Dept. Coordinator	Reviewer	Accreditation Admin
Department_Form	View	View	View	Full
Faculty_Form	View own	View department	View	Full
Accreditation_Criteria_Form	View	View	View	Full
Academic_Year_Form	View	View	View	Full
Evidence_Submission_Form	Add, view own	View department	View	Full
Verification_Form	View own evidence	View department	Add, edit	Full
Accreditation_Dashboard	—	View	View	View

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator under the name **College Accreditation Evidence Management System** on a trial edition (the header of the live application shows “Trial expires in 7 days” at the time the screenshots were taken). It is accessed at the live URL creatorapp.zoho.in/nitinaakash.rk.2024.cse_r965/college-accreditation-evidence-management-system. The six forms were created in the order of the data model – masters first, then transactions – so that each lookup field could point to an existing parent form. Zoho Creator generated a report for each form automatically, and the dashboard page was added last. The resulting navigation menu is visible on the left of every screenshot in this chapter.

5.2 Forms and Data Capture

Figures 5.1 to 5.6 show the six forms in live (data-entry) mode. The design of each form was given in Section 4.3; this section explains the practical purpose of each.

Department_Form (Figure 5.1) stores each department with its code, the coordinator’s e-mail address and an Active/Inactive status. It is the parent of both faculty and evidence records, so it must be filled in first.

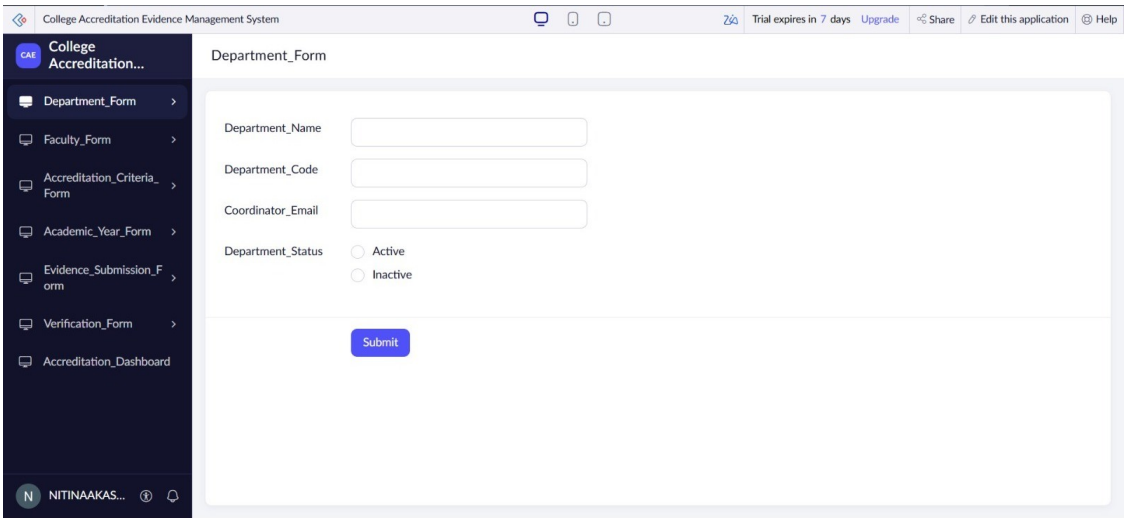


Figure 5.1: Department_Form in Live Mode

Faculty_Form (Figure 5.2) registers each faculty member and links the person to a department through the Department lookup. The Role dropdown records whether the person is a Faculty member, Department Coordinator or Accreditation Admin. Faculty_Name and Faculty_Email are mandatory, as shown by the red asterisks.

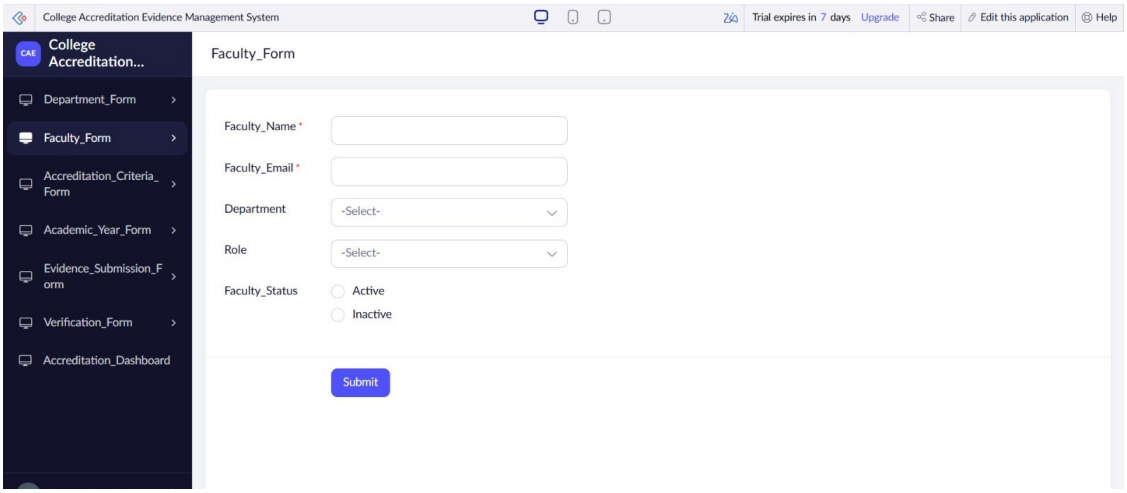


Figure 5.2: Faculty_Form in Live Mode

Accreditation_Criteria_Form (Figure 5.3) defines the criteria against which evidence is collected. The Framework dropdown allows NAAC, NBA or Other, and the Priority radio button helps the accreditation cell decide which criteria to complete first.

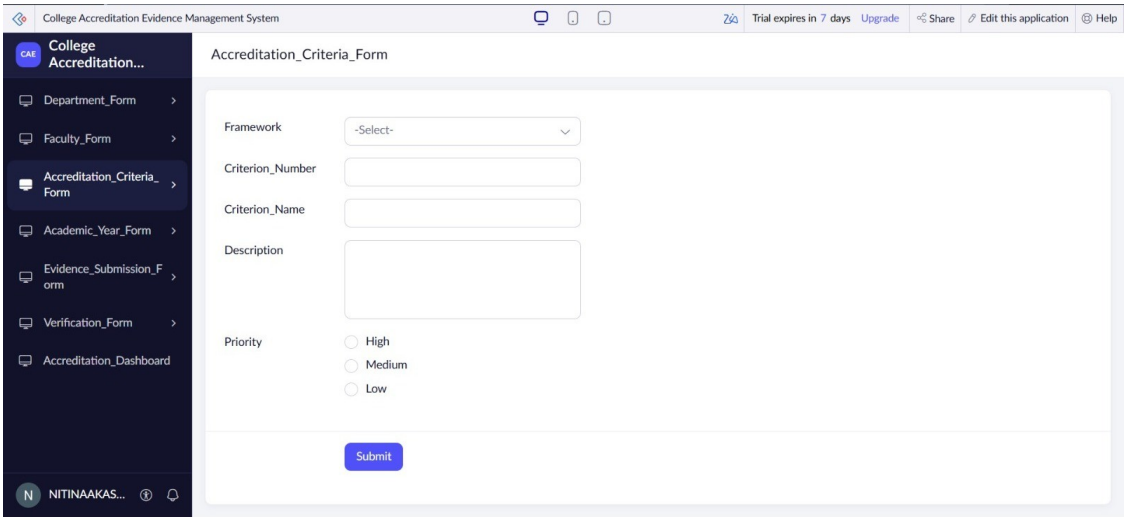


Figure 5.3: Accreditation_Criteria_Form in Live Mode

Academic_Year_Form (Figure 5.4) records each academic year with its start date, end date, evidence submission deadline and Active/Closed status. Two workflows (WF5 and WF6) run on this form.

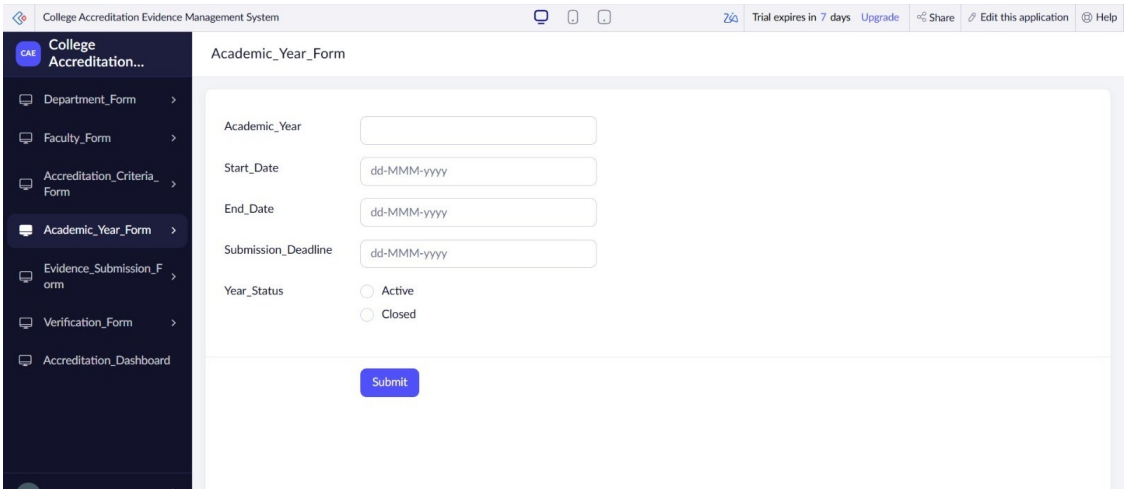


Figure 5.4: Academic_Year_Form in Live Mode

Evidence_Submission_Form (Figure 5.5) is the main transaction form. The user enters a title, selects the department, criterion and academic year from three lookups, and attaches the supporting document through the Evidence_Document upload field.

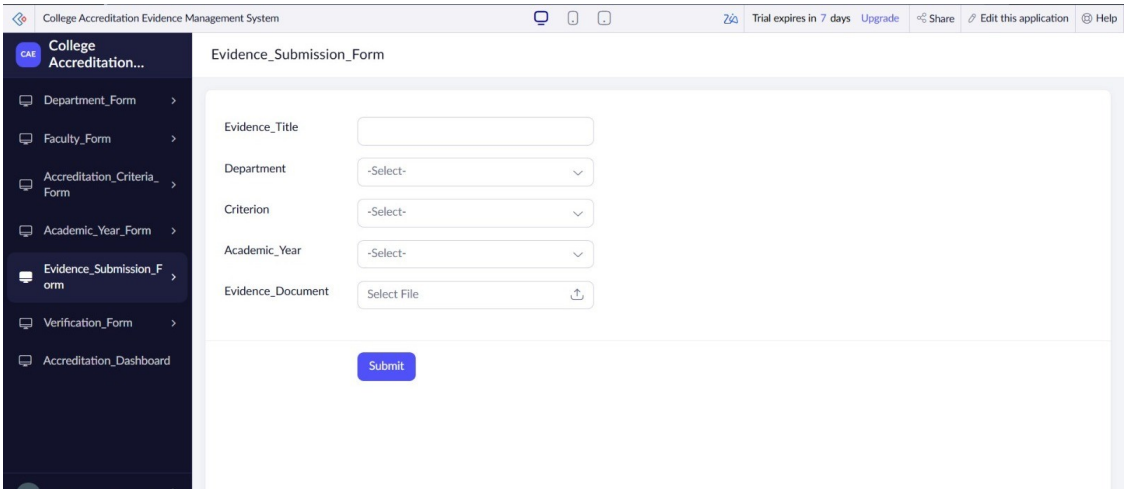


Figure 5.5: Evidence_Submission_Form in Live Mode

Verification_Form (Figure 5.6) records the outcome of a review. The reviewer selects the evidence by its title, selects the reviewing faculty member, chooses a decision and enters comments and a review date. The reviewer lookup is labelled “Faculty_Form” on the form, because its label was left as the name of the parent form; renaming it to “Reviewer” is listed as an improvement in Section 8.4.

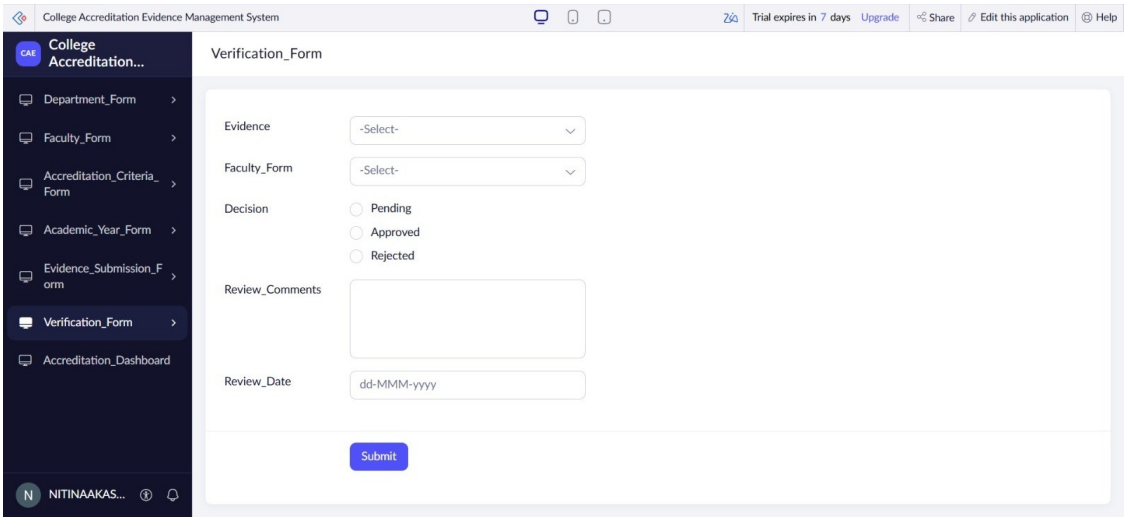


Figure 5.6: Verification_Form in Live Mode

5.3 Reports

Each form has one list report, created by Zoho Creator and shown under the form in the navigation menu. Every report provides a search button, an add-record (+) button, a more-actions menu, column headers with sort and filter options, and a record count at the bottom. Table 5.1 lists the reports and the columns they display, and Figures 5.7 to 5.12 show them with the sample data entered during development.

Table 5.1: Reports Implemented

Report	Columns Displayed	Records
Department_Form Report	Department_ID, Department_Name, Department_Code, Coordinator_Email, Department_Status	3
Faculty_Form Report	Faculty_ID, Faculty_Name, Faculty_Email, Department, Role, Faculty_Status	3
Accreditation_Criteria_Form Report	Criteria_ID, Framework, Criterion_Number, Criterion_Name, Description, Priority	3
Academic_Year_Form Report	Year_ID, Academic_Year, Start_Date, End_Date, Submission_Deadline, Year_Status	3
Evidence_Submission_Form Report	Evidence_ID, Evidence_Title, Department, Criterion, Academic_Year, Evidence_Document	3
Verification_Form Report	Verification_ID, Evidence, Faculty_Form, Decision, Review_Comments, Review_Date	3

The **Department_Form Report** (Figure 5.7) lists three departments – Computer Science and Engineering (CSE), Electronics and Communication (ECE) and Civil Engineering (CIVIL). CSE and ECE are Active and Civil Engineering is Inactive. Coordinator e-mail addresses are shown as clickable mail links.

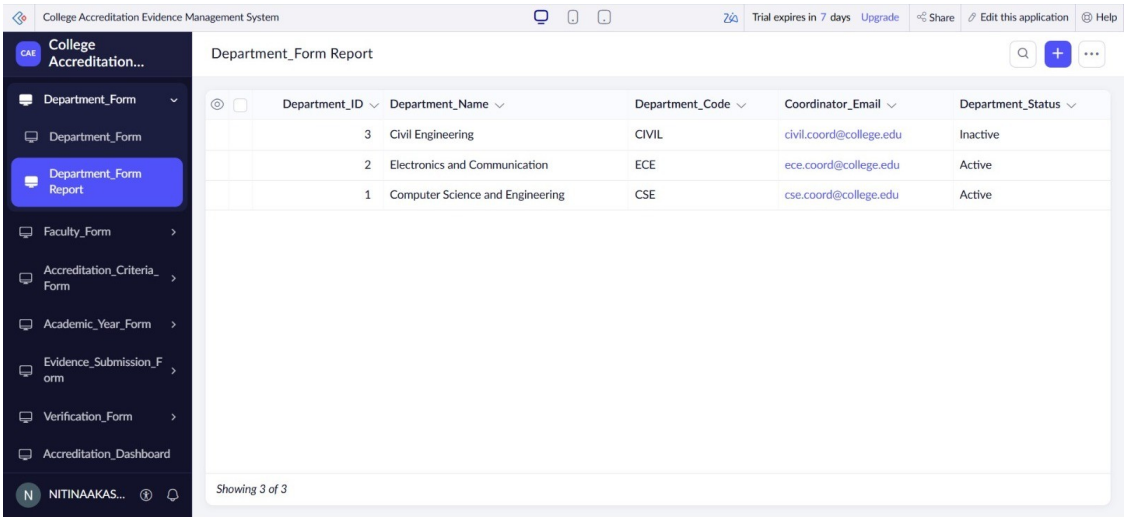


Figure 5.7: Department_Form Report

The **Faculty_Form Report** (Figure 5.8) shows three faculty members with their departments resolved through the lookup: Ananya Rao (CSE, Accreditation Admin), Rahul Kumar (CSE, Faculty) and Priya Sharma (ECE, Department Coordinator), all Active.

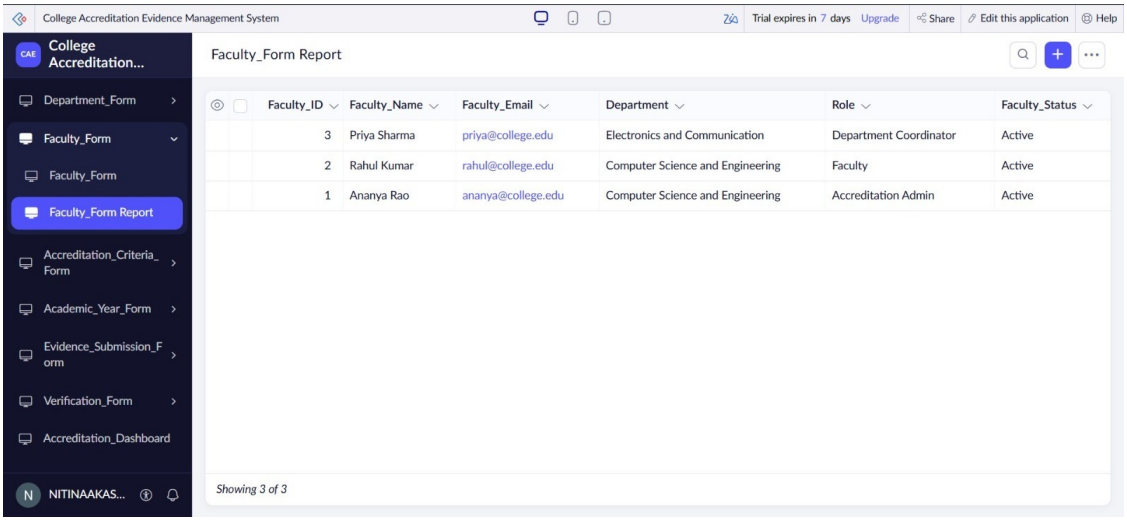


Figure 5.8: Faculty_Form Report

The **Accreditation_Criteria_Form Report** (Figure 5.9) shows three NAAC criteria: 1.1 Curriculum Planning (High), 2.3 Teaching and Learning (High) and 3.2 Research and Innovation (Medium). The Description column is empty for all three.

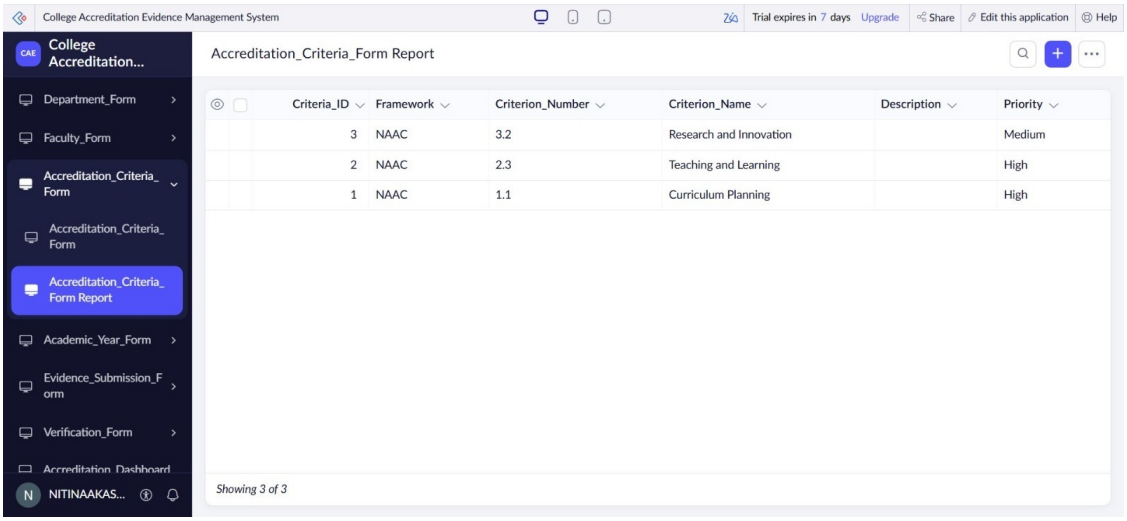


Figure 5.9: Accreditation_Criteria_Form Report

The **Academic_Year_Form Report** (Figure 5.10) shows three academic years. 2024–2025 and 2025–2026 are Closed; 2026–2027 (01-Apr-2026 to 31-Mar-2027, deadline 31-Dec-2026) is Active.

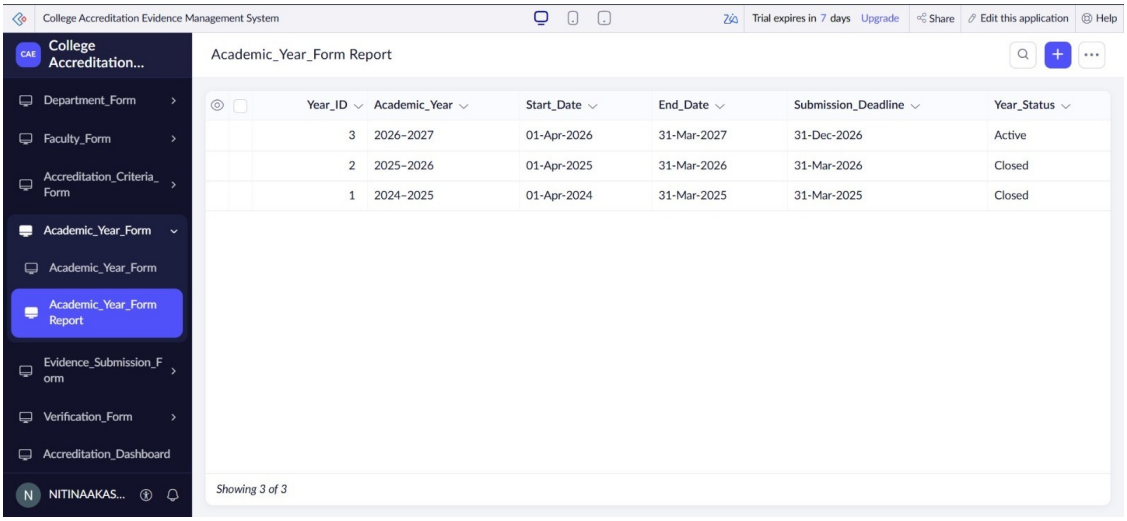


Figure 5.10: Academic_Year_Form Report

The **Evidence_Submission_Form Report** (Figure 5.11) shows three evidence items for 2026–2027: Student Achievement Report (CSE, Curriculum Planning), Faculty Development Records (ECE, Teaching and Learning) and Student Scholarship Records (Civil Engineering, Curriculum Planning). The Evidence_Document column is empty, indicating that files were not attached to these sample records.

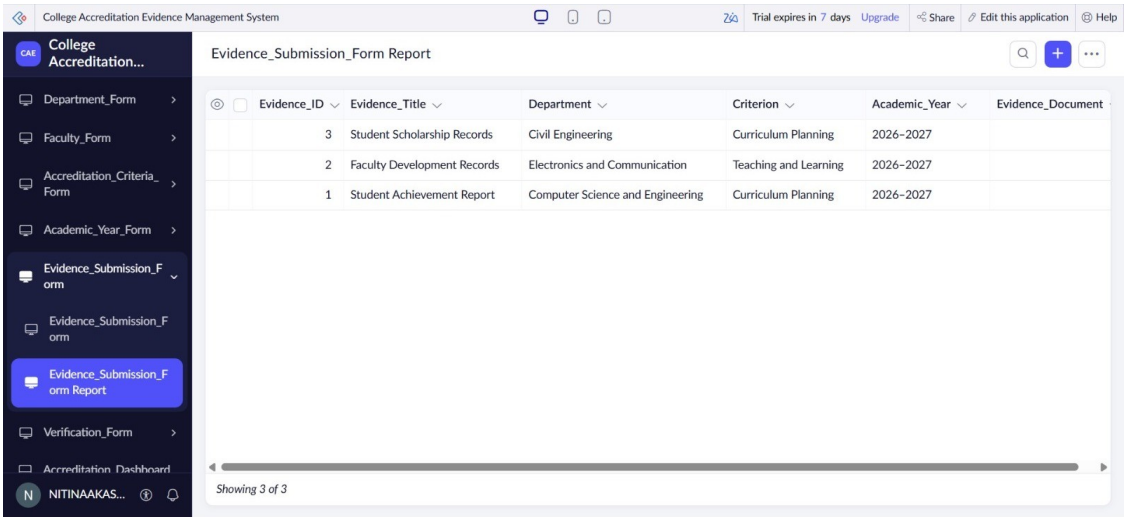


Figure 5.11: Evidence_Submission_Form Report

The **Verification_Form Report** (Figure 5.12) shows three verification records. Verification 3 approves Student Scholarship Records (reviewer Priya Sharma, “Complete evidence”, 13-Oct-2026). Verification 2 keeps Faculty Development Records Pending (“Awaiting review”, no review date). Verification 1 keeps Student Achievement Report Pending although its comment reads “Documents verified” (reviewer Ananya Rao, 10-Oct-2026); this inconsistency in the sample data is discussed in Section 8.4.

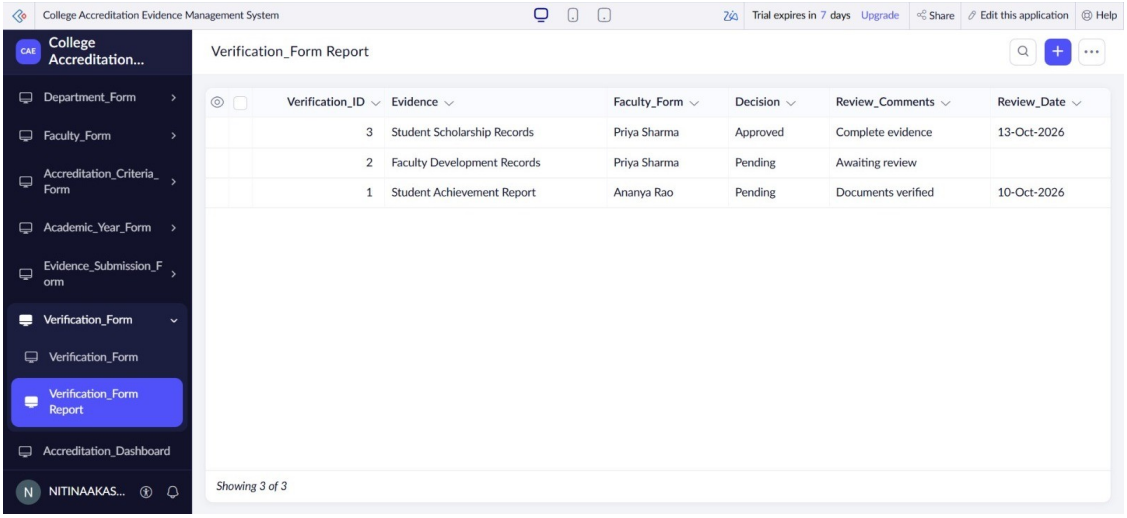


Figure 5.12: Verification_Form Report

The filtered reports considered during design – Active_Departments_Report, Active_Faculty_Report, Pending_Verification_Report and Rejected_Evidence_Report – are not present in the navigation menu and are therefore treated as future enhancements. The same information can currently be obtained by applying column filters in the default reports.

5.4 Pages and Dashboards

The **Accreditation_Dashboard** page (Figure 5.13) gives a single-screen summary of accreditation progress. Its four count cards are computed with aggregate (COUNT) functions over the form data, and its three charts summarise the same data graphically. Table 5.2 lists the components and the values shown at the time of the screenshot.

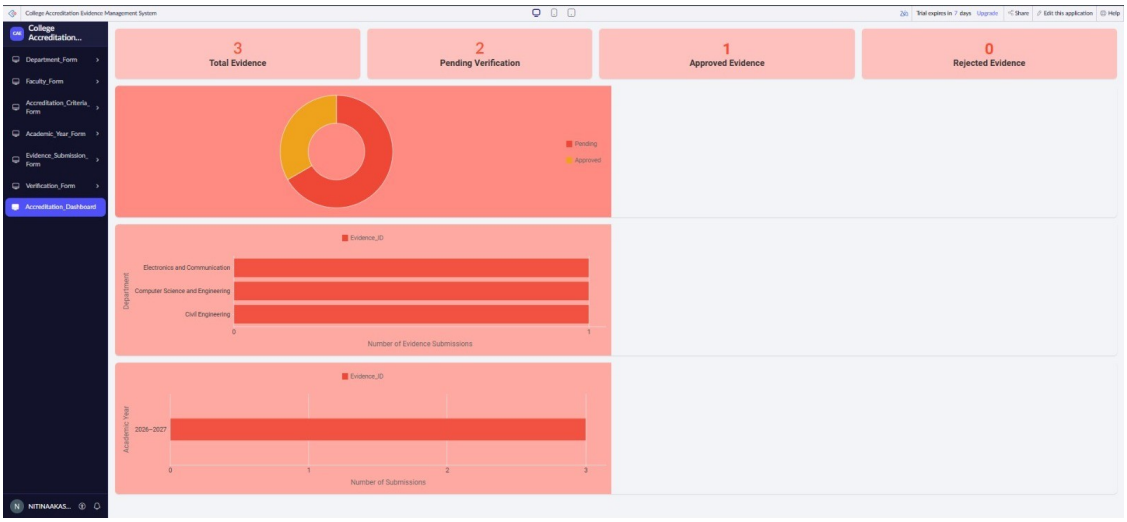


Figure 5.13: Accreditation_Dashboard Page

Table 5.2: Dashboard Components

Component	Type	What It Shows	Value Shown
Total Evidence	Count card	Number of Evidence_Submission_Form records	3
Pending Verification	Count card	Verification records with Decision = Pending	2
Approved Evidence	Count card	Verification records with Decision = Approved	1
Rejected Evidence	Count card	Verification records with Decision = Rejected	0
Evidence status distribution	Donut chart	Share of Pending and Approved decisions	Pending 2, Approved 1
Department-wise evidence	Horizontal bar chart	Evidence count per department (x-axis: Number of Evidence Submissions)	1 each for ECE, CSE, Civil
Academic-year-wise evidence	Horizontal bar chart	Evidence count per academic year (x-axis: Number of Submissions)	2026–2027: 3

The values on the dashboard agree with the reports: three evidence records exist (Figure 5.11), and the three verification records contain two Pending decisions, one Approved decision and no Rejected decision (Figure 5.12). The chart legends read

“Evidence_ID”, the name of the field being counted; renaming the legend to “Evidence count” would make the charts easier to read.

5.5 Deluge Scripting

Deluge is Zoho’s scripting language for business logic inside Creator applications [10]. In this application Deluge is used inside form workflows. Table 5.3 lists the scripts, the evidence available for each and where the code is reproduced. Only code that is visible in the screenshots is reproduced in this report; no code has been reconstructed.

Table 5.3: Deluge Scripts in the Application

Script	Location / Trigger	Purpose	Code Evidence
DS-01 (WF5)	Academic_Year_Form → Created or Edited → Validations on form submission	Set Year_Status to Closed when End_Date is before today	Complete (Figure 5.15)
DS-02 (WF6)	Academic_Year_Form → Created or Edited → Validations on form submission	Send a reminder e-mail when the deadline is 0–7 days away and the year is Active	Lines 1–10 visible (Figure 5.16)
WF1–WF3 scripts	Evidence_Submission_Form, Verification_Form, Faculty_Form	Send notification e-mails	Not captured – awaiting evidence
WF4 script	Verification_Form	Default decision to Pending	Not captured – awaiting evidence

5.5.1 DS-01 – Academic Year Closure (WF5)

The script checks that an end date has been entered and that it is earlier than the current date returned by zoho.currentdate. If both conditions hold, it assigns “Closed” to the Year_Status field of the record being saved. Because it runs during submission, the corrected status is stored with the record.

Code 5.1: Academic year closure script DS-01 (WF5)

```
if(input.End_Date != null && input.End_Date < zoho.currentdate)
{
    input.Year_Status = "Closed";
}
```

5.5.2 DS-02 – Submission Deadline Reminder (WF6)

The script first checks that a submission deadline exists. It then computes the number of days from today to the deadline using daysBetween(). If the deadline is between 0 and 7 days away and the year is Active, it calls sendmail with the Zoho admin user as sender, the developer’s mailbox as recipient and the subject “Accreditation Evidence Submission Deadline Reminder”. The first ten lines, as visible in the workflow builder,

are reproduced below; the remaining lines (message body and closing brackets) were not captured in the screenshot and are marked as awaiting evidence.

Code 5.2: Deadline reminder script DS-02 (WF6, lines 1–10)

```
if(input.Submission_Deadline != null)
{
    daysLeft = daysBetween(zoho.currentdate,input.Submission_Deadline);
    if(daysLeft >= 0 && daysLeft <= 7 && input.Year_Status == "Active")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :"nitinaakash.rk.2024.cse@rajalakshmi.edu.in"
            subject :"Accreditation Evidence Submission Deadline
Reminder"
            // [REMAINING LINES AWAITING IMPLEMENTATION EVIDENCE]
```

The Deluge concepts used are null checks, relational and logical operators, the system variable zoho.currentdate, the date function daysBetween(), assignment to an input field and the sendmail task.

5.6 Workflows and Automation

Figure 5.14 shows the Form workflows tab of the Zoho Creator workflow builder. Four forms appear under “Forms with workflows” – Academic_Year_Form, Evidence_Submission_Form, Faculty_Form and Verification_Form – and two forms appear under “Forms without workflows” – Accreditation_Criteria_Form and Department_Form. With Academic_Year_Form selected, the builder lists WF5_Academic_Year_Closure and WF6_Academic_Year_Deadline_Reminder, each with one action, status Enabled and created on 02-Nov-2026, under the event “Created or Edited → Validations on form submission”.

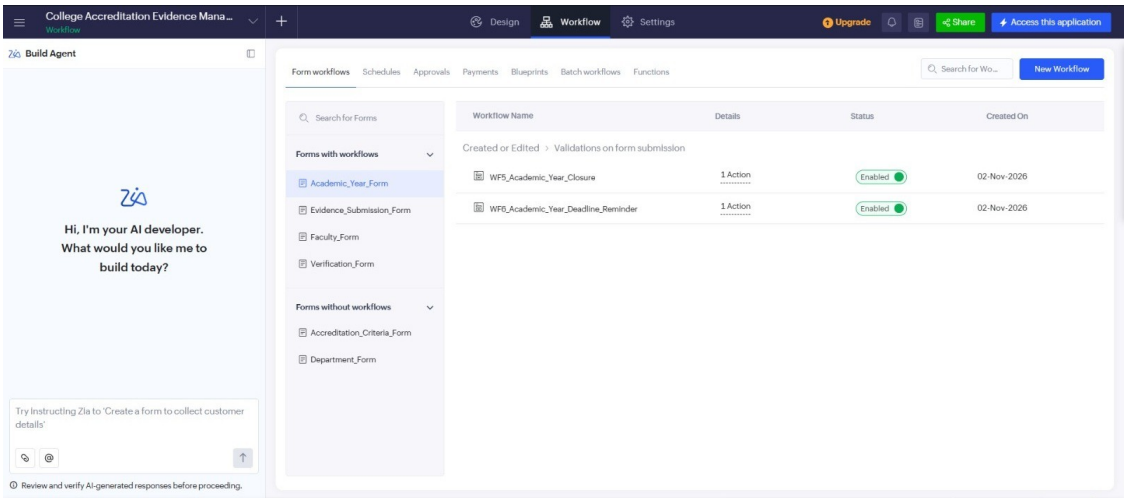


Figure 5.14: Form Workflows List in the Workflow Builder

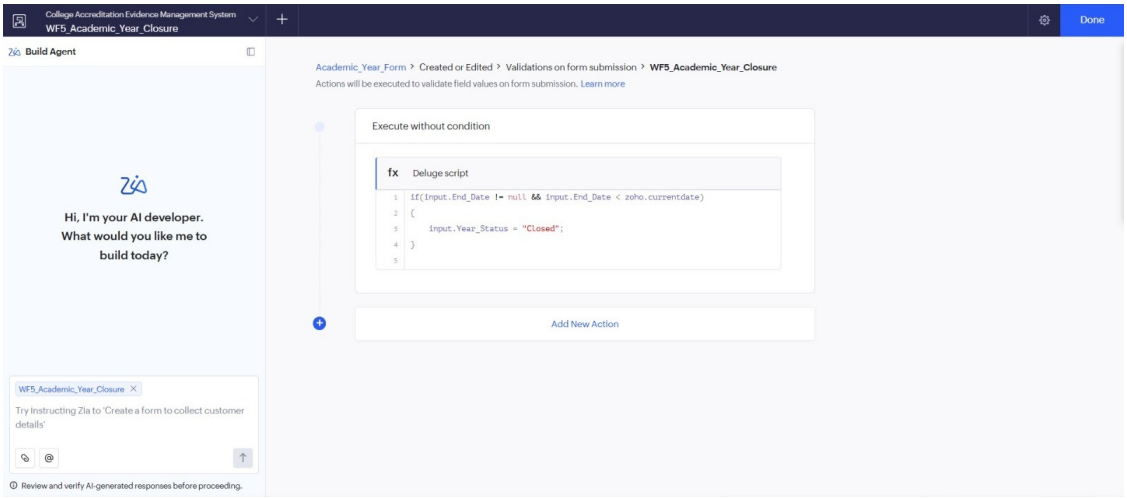


Figure 5.15: WF5_Academic_Year_Closure Configuration

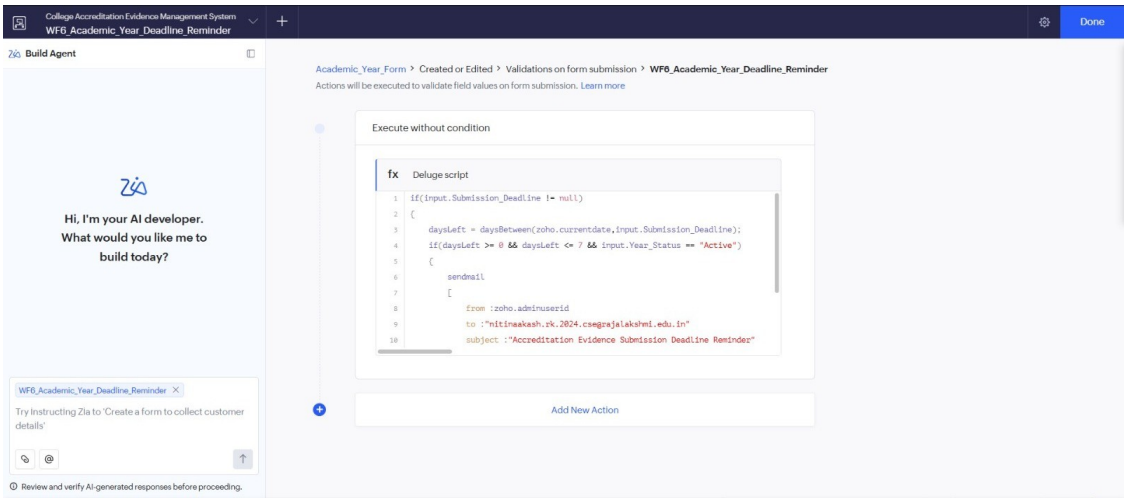


Figure 5.16: WF6_Academic_Year_Deadline_Reminder Configuration

Table 5.4 gives the implementation status of every workflow, distinguishing what is visible in the screenshots from what the developer has confirmed and what still needs evidence. The developer has confirmed that all notification workflows send their e-mail to the developer's personal college mailbox, and that the e-mails were received from the Zoho admin mail address.

Table 5.4: Workflow Implementation Status

WF	Form	Evidence Available	Status
WF1	Evidence_Submission_Form	Form listed under "Forms with workflows"; e-mail receipt confirmed by developer. Trigger and script not captured.	Implemented – developer-confirmed
WF2	Verification_Form	Form listed under "Forms with workflows"; e-mail receipt confirmed by developer. Trigger and script not captured.	Implemented – developer-confirmed

WF	Form	Evidence Available	Status
WF3	Faculty_Form	Form listed under “Forms with workflows”; e-mail receipt confirmed by developer. Trigger and script not captured.	Implemented – developer-confirmed
WF4	Verification_Form	No configuration screenshot.	Awaiting implementation evidence
WF5	Academic_Year_Form	Workflow list and complete script (Figures 5.14, 5.15).	Implemented – on-save logic
WF6	Academic_Year_Form	Workflow list and lines 1–10 of script (Figures 5.14, 5.16); e-mail receipt confirmed by developer.	Implemented – on-save logic

5.6.1 On-Save Workflows versus Scheduled Workflows

Zoho Creator distinguishes between **form workflows**, which run when a user creates, edits or deletes a record, and **schedules**, which run at a configured date and time or at regular intervals without any user action [4]. WF5 and WF6 are form workflows attached to the “Validations on form submission” event. They therefore run only when an Academic_Year_Form record is saved:

- WF5 closes an academic year only if the record is saved after its end date. A year whose end date passes while nobody edits the record remains Active.
- WF6 sends a reminder only if the record is saved within seven days before the deadline. It does not send a reminder automatically each day as the deadline approaches.

The Schedules tab is available in the builder (Figure 5.14), but no schedule has been configured. A daily schedule that fetches all Active academic years and applies the same two rules would make closure and reminders fully automatic; this is listed as a future enhancement. The two records already marked Closed (2024–2025 and 2025–2026) may have been set by WF5 on save or entered as Closed directly; which of these occurred is not evidenced.

5.7 Integrations

The only integration used is e-mail delivery through the Deluge sendmail task, which sends messages from the Zoho admin user address (zoho.adminuserid). No Zoho Cliq, Zoho Analytics, connection, webhook, invokeurl call or REST API integration is used in this version.

Table 5.5: Integration and Platform Feature Status

Feature	Status in this Version	Remarks
E-mail via sendmail	Used	WF1, WF2, WF3 and WF6; delivery confirmed by developer
Other Zoho services (Cliq, Analytics, CRM)	Not used	Future enhancement
Connections / invokeurl / webhooks	Not used	No external system required at this stage
Creator REST API	Not used	Possible future integration with college ERP
Approval process / Blueprint	Not used	Verification is recorded through Verification_Form
Schedules / batch workflows	Not used	See Section 5.6.1

5.8 User Management and Role-Based Access

User roles are represented as data: the Role field of Faculty_Form records each person's responsibility. Zoho Creator user invitations, permission profiles and record-level access criteria are not confirmed for this application, so role-based access control is not claimed. The proposed matrix in Table 4.9 is the basis for configuring it later.

5.9 AI and Analytics Features

The analytics in this version are the dashboard count cards and charts described in Section 5.4. The Zoho Creator builder displays the Zia "Build Agent" AI developer panel beside the workflow editor (Figures 5.14 to 5.16), but no AI field or Zia-generated component is identified as part of the application, and no AI feature is claimed.

5.10 Pro-Code Extension

No pro-code extension (custom widget, Catalyst function or external code) was required. All logic is either configured visually or written in short Deluge scripts inside workflows.

5.11 Mobile and Portal Access

The application header provides preview icons for web, tablet and mobile layouts, and Zoho Creator applications can be opened in the Zoho Creator mobile app. Use of the application on a mobile device, and any customer-portal access for external users, were not tested in this project and are therefore not claimed.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing covered every part of the application: forms, lookups, reports, workflows and the dashboard. Because this report must not overstate results, every test case is assigned one of four evidence statuses rather than a simple Pass/Fail:

- **Evidenced** – the outcome is visible in a screenshot included in this report.
- **Developer-confirmed** – the developer has confirmed the outcome (for example, receipt of an e-mail), but no screenshot of the outcome is included.
- **Observed limitation** – the screenshots show behaviour that differs from the intended design.
- **[TEST RESULT TO BE CONFIRMED]** – the test is defined, but its result has not yet been recorded.

The test levels used were: functional testing of each form (record creation and field behaviour), integration testing of the lookups between forms, workflow testing of the Deluge logic and e-mail delivery, and consistency testing of the dashboard against the underlying reports. Test execution dates were not recorded and are therefore not stated.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected Result	Observed / Evidence	Status
TC-01	FR-01	Create department records	Records saved with auto IDs	3 departments with IDs 1–3 (Fig. 5.7)	Evidenced
TC-02	FR-02	Create faculty records with department lookup	Department name shown for each faculty	Department names resolved (Fig. 5.8)	Evidenced
TC-03	FR-02	Submit Faculty_Form with Faculty_Name blank	Mandatory-field error; record not saved	Field marked mandatory (Fig. 5.2); submission not recorded	[TEST RESULT TO BE CONFIRMED]
TC-04	FR-03	Create criteria records	Records saved with framework and priority	3 NAAC criteria (Fig. 5.9)	Evidenced
TC-05	FR-04	Create academic year records	Records saved with dates in dd-MMM-yyyy	3 years with dates and status (Fig. 5.10)	Evidenced

TC ID	FR Ref.	Test Description	Expected Result	Observed / Evidence	Status
TC-06	FR-05	Submit evidence with department, criterion and year lookups	Lookups show readable names in the report	Names resolved for all three lookups (Fig. 5.11)	Evidenced
TC-07	FR-05	Attach a file in Evidence_Document and open it from the report	File stored and downloadable	Evidence_Document column empty (Fig. 5.11)	[TEST RESULT TO BE CONFIRMED]
TC-08	FR-13	Select an Inactive department for new evidence	Inactive department not offered	Evidence 3 linked to Inactive Civil Engineering (Figs. 5.7, 5.11)	Observed limitation
TC-09	FR-06	Record verification decisions	Decision, comments and date saved	3 verification records (Fig. 5.12)	Evidenced
TC-10	FR-08	WF1: submit evidence and check mailbox	Notification e-mail received	Receipt confirmed by developer	Developer-confirmed
TC-11	FR-08	WF2: submit verification and check mailbox	Decision e-mail received	Receipt confirmed by developer	Developer-confirmed
TC-12	FR-08	WF3: register faculty and check mailbox	Registration e-mail received	Receipt confirmed by developer	Developer-confirmed
TC-13	—	WF4: open Verification_Form	Decision defaults to Pending	Decision unselected in blank form (Fig. 5.6)	[TEST RESULT TO BE CONFIRMED]
TC-14	FR-09	WF5: save a year whose End_Date is past with status Active	Year_Status saved as Closed	Script verified (Fig. 5.15); save result not recorded	[TEST RESULT TO BE CONFIRMED]
TC-15	FR-10	WF6: save an Active year with deadline within 7 days	Reminder e-mail received	Receipt confirmed by developer	Developer-confirmed
TC-16	FR-10	WF6: deadline enters the 7-day window without the record being saved	Reminder sent automatically	No schedule configured (Section 5.6.1)	Observed limitation
TC-17	FR-11	Compare dashboard counts with reports	Counts equal report totals	3 / 2 / 1 / 0 match Figs. 5.11, 5.12	Evidenced
TC-18	FR-11	Compare dashboard charts with reports	Charts equal report groupings	1 per department; 3 for 2026–2027 (Fig. 5.13)	Evidenced

TC ID	FR Ref.	Test Description	Expected Result	Observed / Evidence	Status
TC-19	FR-07	Search and sort records in a report	Matching records listed in order	Search and sort controls visible; use not recorded	[TEST RESULT TO BE CONFIRMED]

In summary, eight test cases are evidenced by screenshots, four are developer-confirmed, two show observed limitations and five still need their results to be recorded. The [TEST RESULT TO BE CONFIRMED] entries should be completed, with a screenshot of each outcome, before final evaluation.

6.3 Debugging Deluge Scripts

The Deluge scripts were written in the workflow builder's script editor, which highlights syntax and reports errors when the script is saved. Two points emerged while reviewing the scripts for this report:

- **Event choice:** both academic-year scripts are attached to "Validations on form submission". This event is suited to checking and correcting input before it is saved; time-based behaviour needs a schedule instead (Section 5.6.1).
- **Hard-coded recipient:** WF6 sends to a fixed address. During development this made delivery easy to verify in one mailbox. For production use, the recipient should be taken from the coordinator e-mail stored in Department_Form, which would require a fetch of the relevant coordinator records.

No Deluge debug log or info-statement output was captured, so none is reproduced here.

6.4 Challenges Faced

In place of a formal defect log, Table 6.2 records the challenges encountered and how they were handled or what remains to be done.

Table 6.2: Challenges Faced

No.	Challenge	Effect	Handling / Next Step
1	Choosing between on-save and scheduled automation for year closure and reminders	Closure and reminders occur only when a record is saved	Kept as on-save logic; daily schedule planned
2	Reviewer lookup label defaulted to the parent form name	Field shows "Faculty_Form" instead of "Reviewer"	Rename the field label
3	Lookup on Department lists all departments	Evidence can be linked to an Inactive department	Add a lookup filter on Department_Status = Active
4	Notification recipient fixed in script	All e-mails go to one mailbox	Fetch coordinator e-mail from Department_Form

No.	Challenge	Effect	Handling / Next Step
5	Sample data entered quickly during development	One Pending record says “Documents verified”; one has no review date; no files attached	Correct the sample records and attach files
6	Trial edition time limit	Application access may end when the trial expires	Capture remaining evidence; move to a permanent plan

6.5 User Acceptance Testing

Formal user acceptance testing with faculty, coordinators or the accreditation cell was not conducted, and no user feedback is reported. A UAT session in which each role performs its main use case (Tables 3.5 to 3.7) and records its observations is recommended before the application is used for an actual accreditation cycle.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Hosting and Access

The application is hosted on Zoho's cloud and is accessed through its live URL on creatorapp.zoho.in; the builder is accessed on creator.zoho.in. Zoho Creator provides Development, Stage and Production environments on supported plans [4], but use of separate environments is not evidenced for this application, so a single live application on the trial edition is described. Table 7.1 summarises the deployment position.

Table 7.1: Deployment Status

Item	Status	Remarks
Application created and accessible in live mode	Done	All forms, reports and dashboard open from the menu
Workflows enabled	Done	WF5 and WF6 shown as Enabled (Fig. 5.14)
Development / Stage / Production environments	Not evidenced	Future enhancement
Users invited with role-based permissions	Not confirmed	Future enhancement (Table 4.9)
Plan	Trial edition	Trial expiry shown in the header; permanent plan needed for continued use

7.2 Performance Considerations

No performance measurements were taken, so no figures are reported. The design nevertheless keeps the application light: each form has no more than six fields, the dashboard uses aggregate counts rather than per-record scripts, and the Deluge scripts perform no database fetches or loops. As the number of evidence records grows over several academic years, the reports can be filtered by academic year to keep each view small.

7.3 Security Features

Table 7.2 separates security features provided by the Zoho Creator platform from controls configured in this application.

Table 7.2: Security Controls

Control	Source	Status
Sign-in required to access the application	Platform	Provided by Zoho accounts
Encrypted connection (HTTPS)	Platform	Application served over https
Hosting and backup of data	Platform	Managed by Zoho

Control	Source	Status
Mandatory fields (Faculty_Name, Faculty_Email)	Application	Configured
Lookup fields restricting values to existing records	Application	Configured
Role-based permission profiles	Application	Not confirmed – future
Record-level access criteria	Application	Not confirmed – future
Audit trail review	Application	Not evidenced – future

7.4 Governance and Data Quality

For an accreditation system, governance is mainly about the trustworthiness of the evidence. The present version supports governance through auto-numbered IDs, linked master data and recorded review dates. The review of sample data for this report also showed where governance should be strengthened: decisions should agree with comments, a review date should be mandatory whenever a decision other than Pending is recorded, criteria descriptions should be filled in, inactive departments should be excluded from new submissions, and every evidence record should carry its document. These rules can be enforced through mandatory-field settings, lookup filters and validation scripts in a future version.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The output of the application is best understood by following one evidence item through the system using the actual records. Table 8.1 traces the Student Scholarship Records item from master data to dashboard, with the figure that shows each step.

Table 8.1: End-to-End Scenario Using the Recorded Data

Step	Action	Record in the Application	Figure
1	Admin defines the department	Civil Engineering, code CIVIL, civil.coord@college.edu (now Inactive)	5.7
2	Admin defines the criterion	NAAC 1.1 Curriculum Planning, priority High	5.9
3	Admin defines the academic year	2026–2027, Active, deadline 31-Dec-2026	5.10
4	Faculty submits evidence	Evidence 3, Student Scholarship Records, Civil Engineering, Curriculum Planning, 2026–2027	5.11
5	Notification is sent	WF1 e-mail received (developer-confirmed)	—
6	Reviewer records decision	Verification 3, Priya Sharma, Approved, “Complete evidence”, 13-Oct-2026	5.12
7	Decision notification is sent	WF2 e-mail received (developer-confirmed)	—
8	Dashboard reflects progress	Approved Evidence = 1; Civil Engineering bar = 1	5.13

The scenario shows that the core path – classify, submit, verify and monitor – works with the lookups resolving correctly at every step. It also illustrates one limitation: the evidence was accepted for a department that is now Inactive.

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Section 1.3)	Outcome	Status
Data model and six forms	Six forms with lookups built as designed (Figs. 5.1–5.6)	Achieved
Link evidence and verification through lookups	Six lookups resolve to readable names in reports	Achieved
List reports for every form	Six reports with search, sort and add (Figs. 5.7–5.12)	Achieved
Automate notifications, closure and reminders	E-mail workflows confirmed; closure and reminder run on save only	Partially achieved
Dashboard with counts and charts	Four count cards and three charts (Fig. 5.13)	Achieved

Objective (Section 1.3)	Outcome	Status
Reduce paperwork and improve organisation	Evidence classified by department, criterion and year in one place	Achieved for the sample data; not measured in live use

8.3 Comparison with the Existing System

Table 8.3: Existing System versus Proposed Application

Aspect	Existing System	Proposed Application
Storage of evidence	E-mail, local folders, paper files	One application with an upload field per evidence record
Classification	Inconsistent naming of files	Department, criterion and academic year chosen from lookups
Verification record	Informal notes, if any	Decision, reviewer, comments and date in Verification_Form
Progress visibility	Manual counting	Dashboard counts and charts
Notifications	Manual e-mails and calls	Form workflows send e-mails on submission and decision
Deadline handling	Manual reminders	On-save reminder (scheduled reminder planned)
Continuity when staff change	Knowledge held by individuals	Records held centrally with IDs

8.4 Observed Limitations of the Current Build

The application is documented as it stands. The following limitations were identified from the screenshots and are recorded so that they can be corrected:

1. **Reviewer field label:** the reviewer lookup in Verification_Form and its report column are labelled “Faculty_Form” instead of “Reviewer”.
2. **Inconsistent verification samples:** Verification 1 is Pending although its comment says “Documents verified”, and Verification 2 has no Review_Date.
3. **Empty evidence files:** none of the three evidence records has a document attached, so file upload and retrieval are not yet demonstrated.
4. **Missing criteria descriptions:** the Description field is empty for all three criteria.
5. **Inactive departments selectable:** the Department lookup does not filter by status, and Evidence 3 is linked to the Inactive Civil Engineering department.
6. **On-save automation only:** WF5 and WF6 run only when an academic-year record is saved; there is no scheduled closure or reminder.

7. **Fixed e-mail recipient:** notifications go to the developer's mailbox rather than to the department coordinator.
8. **No submitter field:** Evidence_Submission_Form does not record which faculty member submitted the evidence.
9. **Filtered reports not built:** the four planned filtered reports are not present.
10. **Roles not enforced:** roles are stored as values; permission profiles are not confirmed.
11. **Few mandatory fields:** only Faculty_Name and Faculty_Email are mandatory.
12. **Chart legends:** the bar-chart legends read "Evidence_ID".

8.5 Limitations of the Low-Code Approach

- **Platform dependence:** the application, its data and its scripts run only on Zoho Creator, a form of vendor lock-in noted in the literature [7].
- **Plan limits:** the trial edition limits how long the application can be used without a paid plan.
- **Bounded customisation:** layouts, charts and behaviour are limited to what the platform offers, although Deluge covers most business rules [9].
- **Debugging visibility:** errors in workflow scripts are less visible than in a conventional development environment with step-through debugging.

8.6 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of this application was built in **no-code** mode: forms, lookups, reports and the dashboard were configured entirely through the visual builders. **Low-code** was used where rules were needed – the Deluge scripts for academic-year closure, deadline reminders and e-mail notifications. **Pro-code** was not needed. The experience shows that an accreditation evidence system sits comfortably in the low-code range; pro-code would become relevant only for deep integration with a college ERP or a custom document viewer.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The College Accreditation Evidence Management System brings accreditation evidence into one Zoho Creator application. Six forms with lookup relationships capture departments, faculty, accreditation criteria, academic years, evidence submissions and verification decisions; six list reports present the records; form workflows send e-mail notifications whose delivery has been confirmed by the developer; Deluge logic updates academic-year status and sends a deadline reminder when a record is saved; and the Accreditation_Dashboard summarises evidence and verification through count cards and charts whose values agree with the underlying reports.

The project met its central objective of classifying evidence by department, criterion and academic year and recording its verification in a structured, traceable way. Some objectives are partially met, notably fully automatic closure and reminders, and several limitations were identified honestly from the screenshots. These form a clear and practical roadmap for the next version.

9.2 Future Enhancements

- Add daily **scheduled workflows** for automatic academic-year closure and deadline reminders.
- Send notifications to the **department coordinator** by fetching Coordinator_Email from Department_Form instead of a fixed address.
- Build the **filtered reports** Active_Departments_Report, Active_Faculty_Report, Pending_Verification_Report and Rejected_Evidence_Report, and add the pending report to the dashboard.
- Add a **lookup filter** so that only Active departments, faculty and academic years can be selected.
- Rename the reviewer field to **Reviewer**, add a **Submitted_By** field to evidence, and make the key fields mandatory.
- Configure **role-based permission profiles** according to Table 4.9.
- Implement WF4 and add validation so that a decision other than Pending requires a review date and comments.
- Use an **approval process or Blueprint** for multi-level verification by coordinator and accreditation cell.

- Use **Development, Stage and Production** environments and the audit trail before live use.
- Integrate with **Zoho Analytics** for criterion-wise completion analysis, and with **Zoho Cliq** for team alerts.
- Test and publish the application for the **Zoho Creator mobile app**.
- Provide criterion-wise **evidence compilation exports** for the self-study report.

9.3 Learning Outcomes and Career Relevance

The project provided practical experience in modelling a real institutional process as related data entities, building forms and lookups, writing Deluge scripts with date functions and the sendmail task, and summarising data on a dashboard. It also highlighted professional practices that matter beyond the platform: choosing the correct trigger for automation, avoiding hard-coded values, keeping sample data consistent, and reporting implementation status honestly with evidence. These skills are directly relevant to roles in business application development, process automation and low-code consulting.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Each feature is marked Y (implemented and evidenced), P (partially implemented or developer-confirmed without a screenshot) or N (not implemented in this version), with its location and the report section and figure that support it.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Section	Figure
I	Application created and menu organised	Y	Seven menu items: six form sections and the dashboard	5.1	4.4
II	Forms with varied field types	Y	Auto Number, Single Line, Multi Line, Email, Date, Radio, Dropdown, Lookup, File Upload	4.3, 5.2	5.1–5.6
II	Lookup fields between forms	Y	Six lookups across Faculty, Evidence and Verification forms	4.2	4.2
II	Subform (parent–child)	N	Not required by the design	—	—
II	Field validations / mandatory fields	P	Faculty_Name and Faculty_Email mandatory	4.3	5.2
II	List reports	Y	Six default reports	5.3	5.7–5.12
II	Filtered / custom reports	N	Planned filtered reports not built	5.3	—
II	Kanban / Calendar / Pivot reports	N	Not used	—	—
II	Page / dashboard	Y	Accreditation_Dashboard with COUNT cards and three charts	5.4	5.13
II	Deluge – form events	Y	WF5 and WF6 on Validations on form submission	5.5	5.15, 5.16
II	E-mail notifications	P	WF1, WF2, WF3, WF6 (developer-confirmed delivery)	5.6	5.16
II	Form workflows	Y	Workflows on four forms	5.6	5.14
II	Scheduled workflow	N	Schedules tab not used	5.6.1	—
II	Approval process / Blueprint	N	Not used	5.7	—
II	Testing evidence	P	Test cases with evidence status	6.2	—
III	Integration with a Zoho service	P	E-mail through sendmail only	5.7	—
III	Connections / REST API / webhooks	N	Not used	5.7	—
III	Users, roles and permission profiles	N	Role stored as a field value only	5.8	—
III	Zia / AI feature	N	Not used	5.9	—

Unit	Zoho Creator Feature	Done	Where in the Application	Section	Figure
III	Analytics (charts)	Y	Donut and bar charts on the dashboard	5.4	5.13
IV	Performance optimisation measured	N	Considerations only; no measurements	7.2	—
IV	Deployment across environments	N	Single live application on trial edition	7.1	—
IV	Security controls	P	Platform sign-in and HTTPS; app-level permissions not confirmed	7.3	—
IV	Governance (audit trail)	N	Not evidenced	7.4	—
V	Pro-code extension	N	Not required	5.10	—
V	Low-code limitations and spectrum analysis	Y	Discussed	8.5, 8.6	—
—	Mobile / portal access	N	Not tested	5.11	—

APPENDIX B – COURSE OUTCOME MAPPING

The table shows where each Course Outcome of CB23F35 is evidenced in this report and application. The attainment levels (1 – Low, 2 – Medium, 3 – High) are to be entered by the student and may be revised by the examiner.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.5, 8.6, 9.3		

APPENDIX C – AVAILABLE DELUGE SOURCE CODE

Only Deluge code that is visible in the application screenshots is reproduced. Code that was not captured is marked as awaiting implementation evidence and has not been reconstructed.

C.1 WF5_Academic_Year_Closure (DS-01) – complete

```
// Form    : Academic_Year_Form
// Event   : Created or Edited -> Validations on form submission
// Purpose : Close an academic year whose end date has passed
if(input.End_Date != null && input.End_Date < zoho.currentdate)
{
    input.Year_Status = "Closed";
}
```

C.2 WF6_Academic_Year_Deadline_Reminder (DS-02) – lines 1–10

```
// Form    : Academic_Year_Form
// Event   : Created or Edited -> Validations on form submission
// Purpose : Reminder e-mail when the deadline is 0-7 days away
if(input.Submission_Deadline != null)
{
    daysLeft = daysBetween(zoho.currentdate,input.Submission_Deadline);
    if(daysLeft >= 0 && daysLeft <= 7 && input.Year_Status == "Active")
    {
        sendmail
        [
            from :zoho.adminuserid
            to :"nitinaakash.rk.2024.cse@rajalakshmi.edu.in"
            subject :"Accreditation Evidence Submission Deadline
Reminder"
            // [REMAINING LINES AWAITING IMPLEMENTATION EVIDENCE]
```

C.3 WF1 to WF4

Table C.1: Scripts Awaiting Implementation Evidence

Workflow	Form	Code Status
WF1 New Evidence Submission Notification	Evidence_Submission_Form	[AWAITING IMPLEMENTATION EVIDENCE]
WF2 Verification Decision Notification	Verification_Form	[AWAITING IMPLEMENTATION EVIDENCE]
WF3 New Faculty Registration Notification	Faculty_Form	[AWAITING IMPLEMENTATION EVIDENCE]
WF4 Default Verification Decision	Verification_Form	[AWAITING IMPLEMENTATION EVIDENCE]

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the live URL creatorapp.zoho.in/nitinaakash.rk.2024.cse_r965/college-accreditation-evidence-management-system in a web browser.
2. Sign in with the Zoho account that has access to the application.
3. Use the left menu to open a form, its report or the Accreditation_Dashboard.

D.2 Instructions for the Accreditation Admin

1. Open Department_Form, enter the department name, code and coordinator e-mail, choose Active, and click Submit.
2. Open Faculty_Form, enter the name and e-mail (both mandatory), select the department and role, choose the status, and click Submit.
3. Open Accreditation_Criteria_Form, choose the framework, enter the criterion number, name and description, choose the priority, and click Submit.
4. Open Academic_Year_Form, enter the year, start date, end date and submission deadline, choose the status, and click Submit.
5. To change a record, open the relevant report, select the record and edit it.

D.3 Instructions for Faculty

1. Open Evidence_Submission_Form.
2. Enter a clear Evidence_Title.
3. Select your Department, the Criterion and the Academic_Year.
4. Click Select File in Evidence_Document and attach the supporting document.
5. Click Submit, then confirm the record in Evidence_Submission_Form Report.

D.4 Instructions for the Reviewer

1. Open Verification_Form.
2. Select the Evidence by its title.
3. In the field labelled Faculty_Form, select your own name as reviewer.
4. Choose Pending, Approved or Rejected; enter Review_Comments and Review_Date.
5. Click Submit, then confirm the record in Verification_Form Report.

D.5 Instructions for the Department Coordinator

1. Open Evidence_Submission_Form Report and filter by your department to see submissions.
2. Open Verification_Form Report to see decisions on your department’s evidence.

3. Open Accreditation_Dashboard for the overall counts and charts.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	NITIN AAKASH RK	**Register No.**	240701370
Project Title	College Accreditation Evidence Management System	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.

Level	% of Max. Marks	Descriptor
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	NITIN AAKASH RK	**Register No.**	240701370
Department / Section	Computer Science and Engineering	**Date of Viva**	11.11.2026
Application URL Verified		**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_

_____

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INTERNAL EXAMINER
Name: _____

EXTERNAL EXAMINER
Name: _____